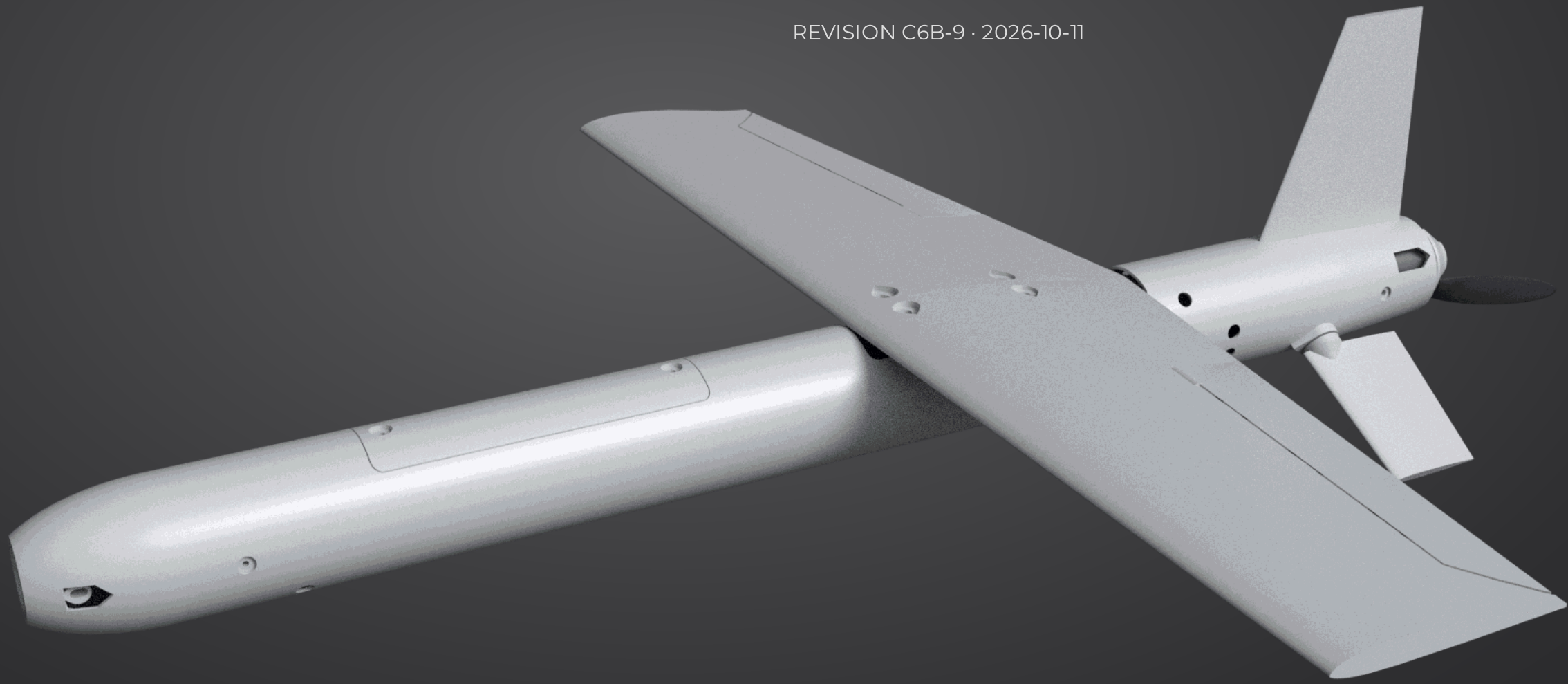
SKYG

SKYG SPEED 90B

USER MANUAL

REVISION C6B-9 · 2026-10-11



PACKAGE CONTENTS

Manual

SkyG Speed 90B user manual (this document).

STL

Every printed part, in millimetres, ready for the slicer (print orientation: pages in this manual).

STEP

Every printed part and the full assembly, for editing in any CAD program.

Wiring

The wiring diagram as a separate large page.

ArduPilot

Setup document and a starting .param file for the SpeedyBee F405 WING MINI.

Viewer

Interactive 3D model (open the HTML file in a browser): every part exploded and in place, with notes.

Engineering

Design record, parts list (BOM.csv), test plan and verification report.

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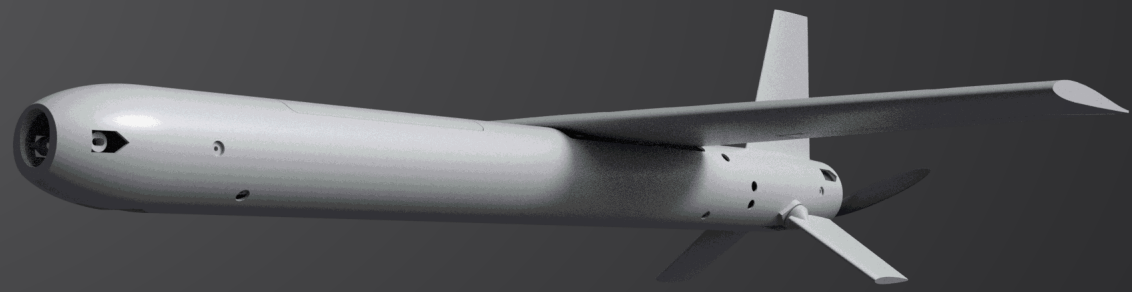
Flight test and speed stages 53

SKYG SPEED 90B

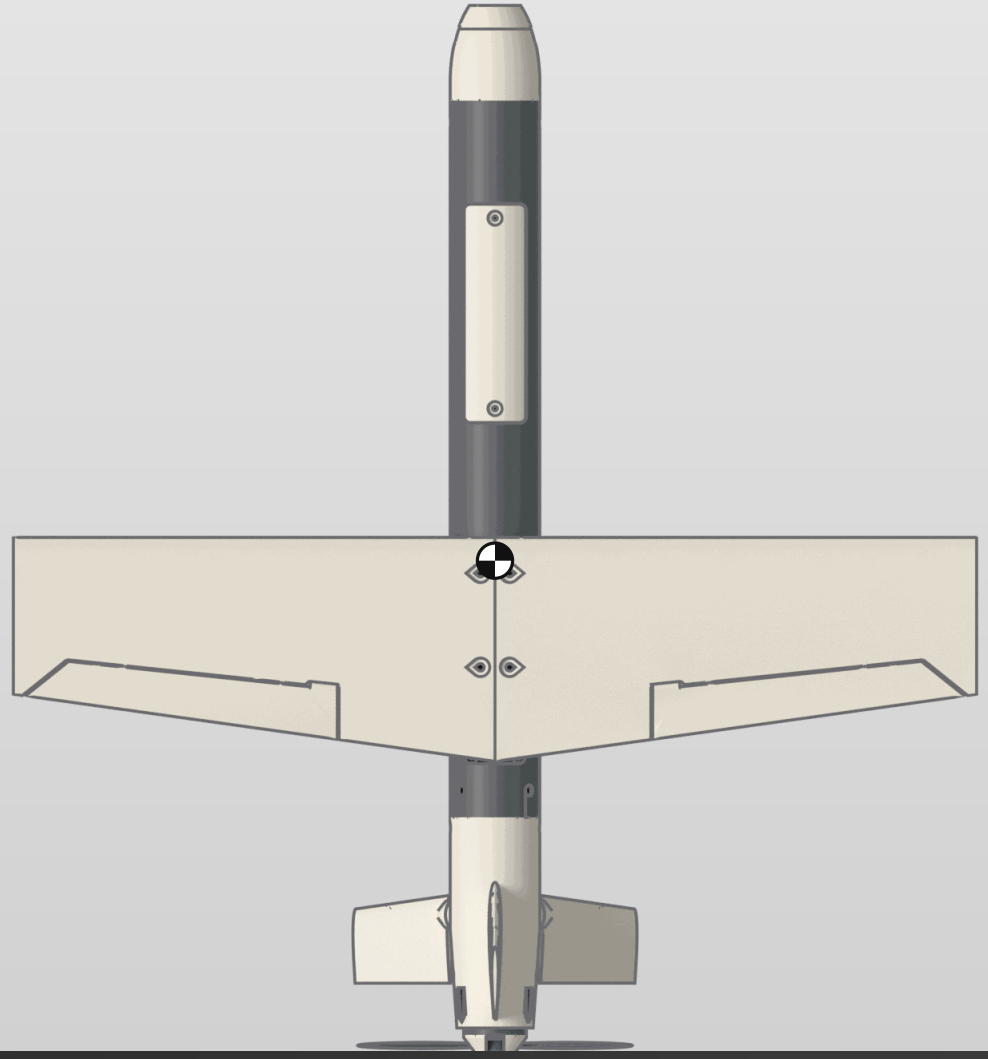
SkyG Speed 90B is a compact, hand-launched speed aircraft: a 3D-printed fuselage in three plug-together modules and a separate camera nose, a 430 mm straight wing on a carbon spar, and an all-moving canted tail driven by two micro servos. A single pusher motor sits in the tail cone behind the tail.

The airframe is printed in foaming PLA Aero on a 0.4 mm nozzle; steel and carbon parts carry the concentrated loads (spar, aileron hinge pins and torque rods, tail shafts). Six interchangeable camera noses take a micro FPV camera, a Raspberry Pi camera or a DJI O4 Air Unit, each with an open or a clear-dome face.

It is built for straight-line speed: about 87 m/s finished with the planned rpm governor, and about 63 m/s until that governor is fitted, flown up through never-exceed stages.



GENERAL AIRCRAFT DATA



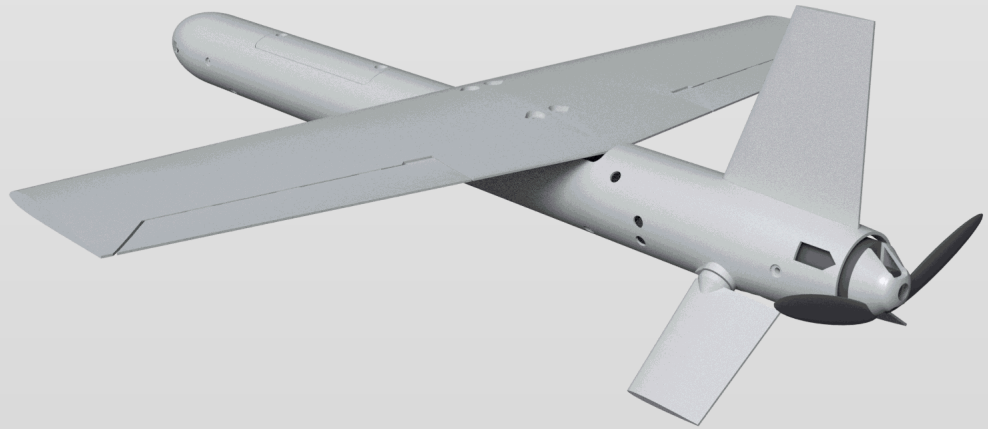
Wingspan	430 mm
Wing area	3.65 dm²
Length	471 mm
Centre of gravity	10 mm behind the wing leading edge at the root
All-up weight	516 g (micro-camera nose)
Stall speed	13.0 m/s
Top speed (model)	86–87 m/s with the governor
Airfoil	MH45, ailerons 3° reflex
Root / tip chord	100 / 70 mm
MAC	86 mm
Aspect ratio	5.1
Wing loading	104 g/dm²

AERODYNAMIC DESIGN

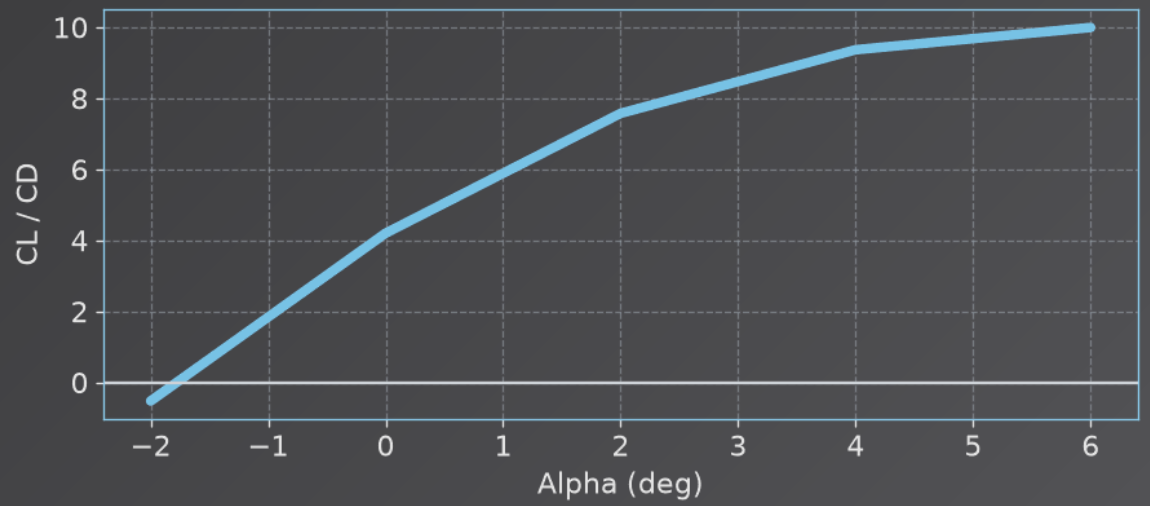
The wing uses the MH45 section, a low-moment reflexed profile, with both ailerons rigged 3° trailing edge up. The design CG is at 12 % of the mean chord, 6.5 % static margin ahead of the neutral point (x 261 mm from the nose datum): stable without a tail download penalty at speed.

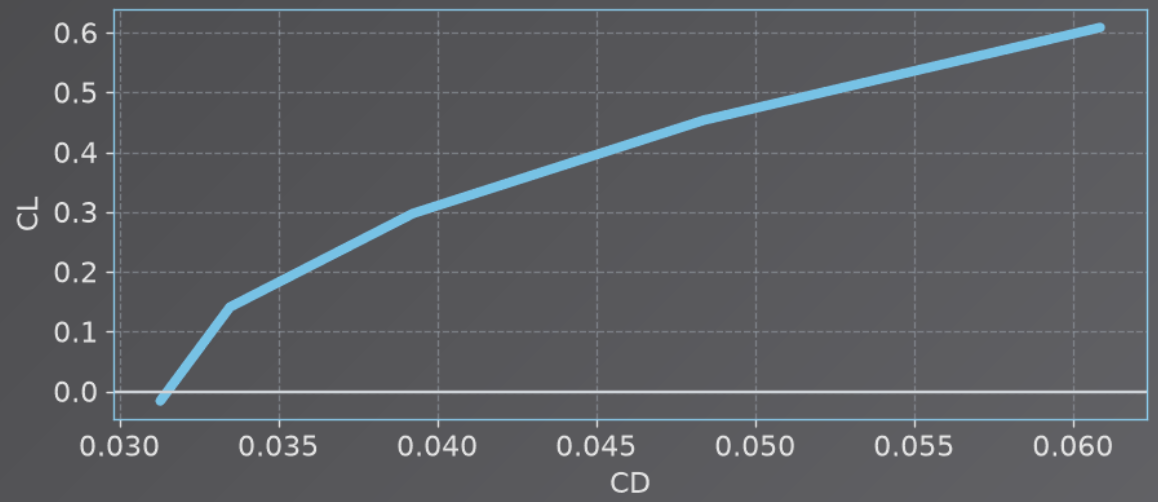
Pitch and yaw come from two all-moving tail panels at 35° anhedral, mixed like a V-tail, with a fixed fin on top. The Dutch roll is lightly damped at low speed, so the flight controller's yaw damper is part of the setup.

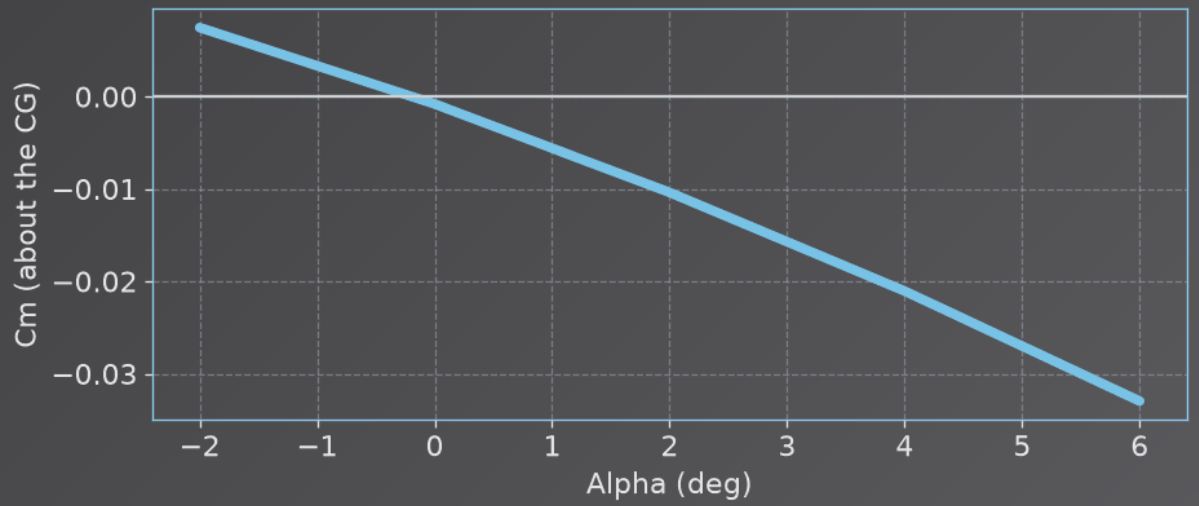
Flutter of the wing, ailerons and tail panels was analysed (strip theory with the CAD-derived stiffness and masses): the lowest predicted instability is 137 m/s, above 1.2 × the 90 m/s top stage. The test plan opens the speed envelope in stages and measures the stiffness the margin depends on.

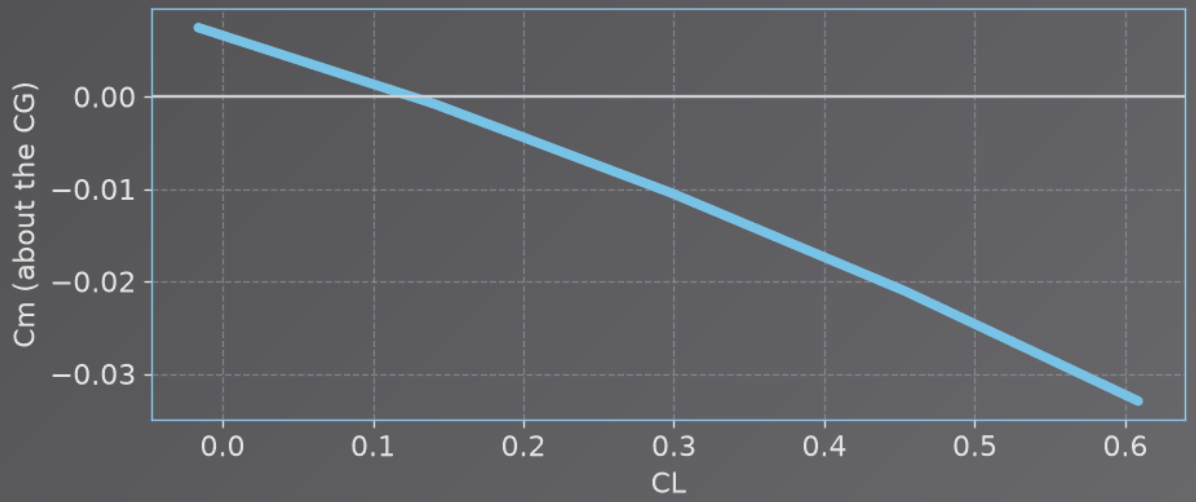


AERODYNAMIC CHARACTERISTICS









Whole aircraft, VSPAERO vortex lattice with the fuselage panels; best L/D 10.0 at 6.0°; Cm about the design CG.

PERFORMANCE AND SPEED STAGES

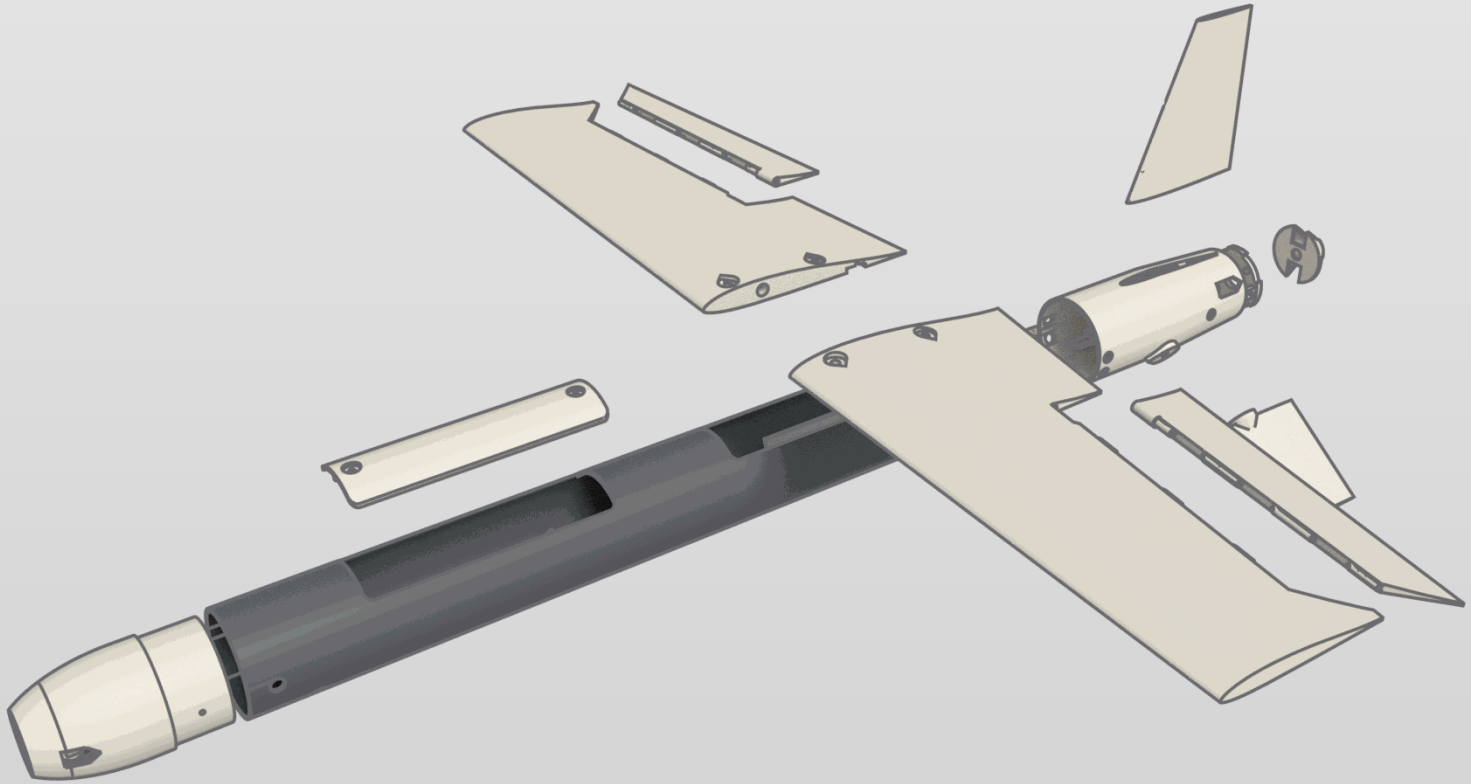
Motor	2207 2550KV 4S racing motor
Propeller	APC 5x7.5E (LP05075E)
Battery	Tattu R-Line 4S 650 75C
Static thrust	1064 g at 51 A (thrust / weight 2.06)
Throttle cap (prop rpm rating)	70 %
Top speed	63–63 m/s now; 86–87 m/s with the governor
Full-throttle time	about 0.7 min per pack (bursts of ≤ 10 s)

Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	28 %	24 m/s
35 m/s	32 %	28 m/s
40 m/s	36 %	32 m/s
45 m/s	41 %	37 m/s
50 m/s	45 %	41 m/s
55 m/s	50 %	45 m/s
60 m/s	55 %	50 m/s
65 m/s	59 %	53 m/s
70 m/s	64 %	58 m/s
75 m/s	69 %	62 m/s
80 m/s	70 % (needs the governor)	63 m/s
85 m/s	70 % (needs the governor)	63 m/s
90 m/s	70 % (needs the governor)	63 m/s

No airspeed sensor: each stage is held by a throttle limit on a transmitter switch (LAUNCH = full cap for the throw and climb-out only, STAGE = the limit). Level runs only; speed from GPS two-way passes.

FILES

Every printed part is supplied as STL (for printing) and STEP (for editing), plus the complete assembly as one STEP file. Left and right wing halves, ailerons and tail panels are separate files because they are mirror images. The interactive viewer in the package shows every part, purchased item and cable in place.



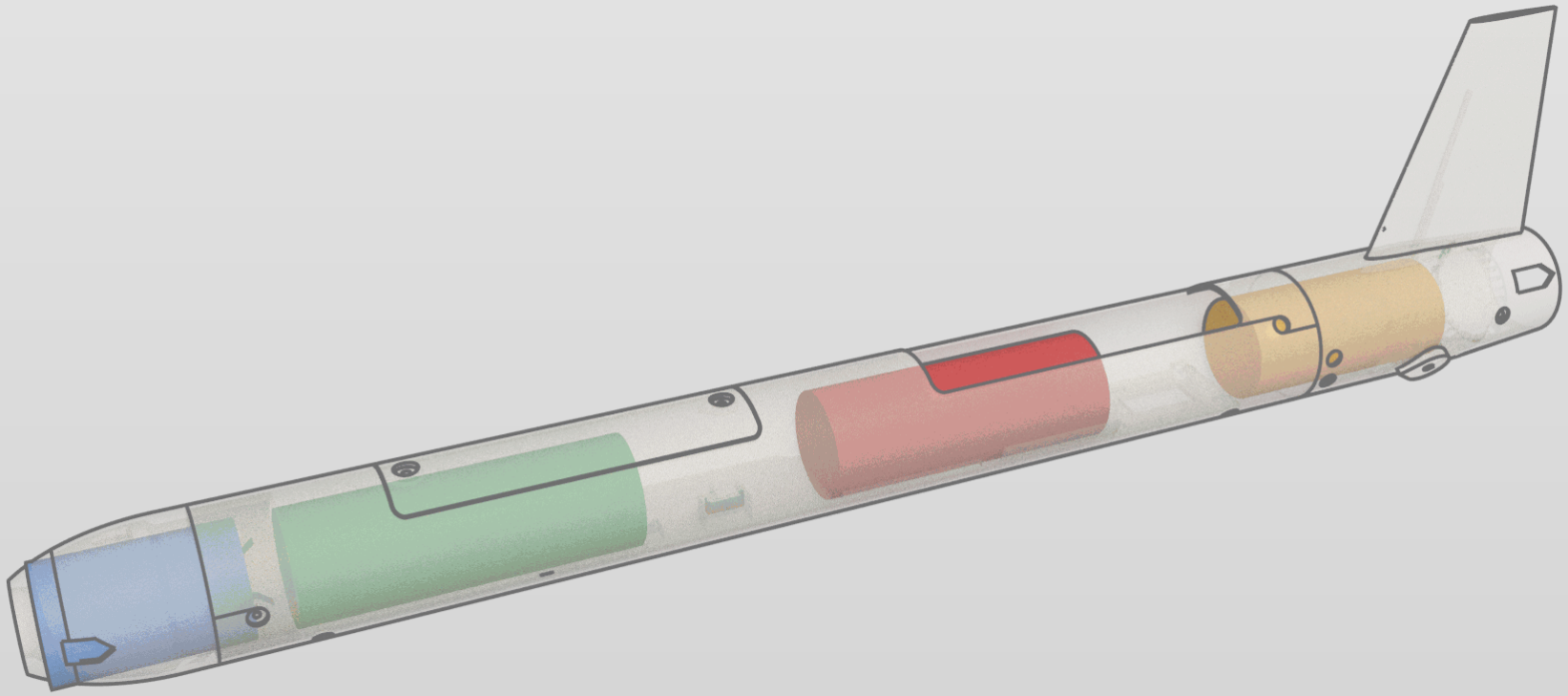
COMPARTMENT LAYOUT

BATTERY BAY
4S 650 pack, 60 × 31 × 21 mm max; slides x 108–128 for balance; rear stop at x 158

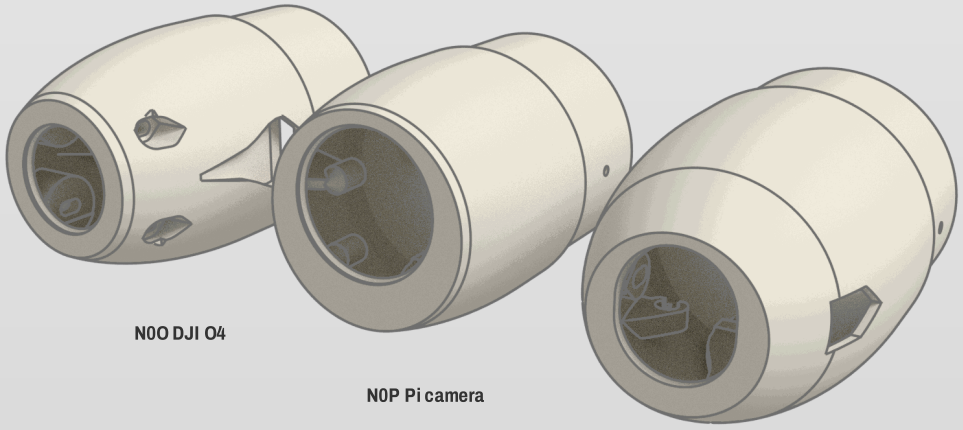
WING BAY
Flight controller stack with the ESC on top, receiver, aileron servo; reached with the wing off

TAIL MODULE
Two tail servos, pushrods, horns and the motor; plugs off at the MR30 and servo plugs

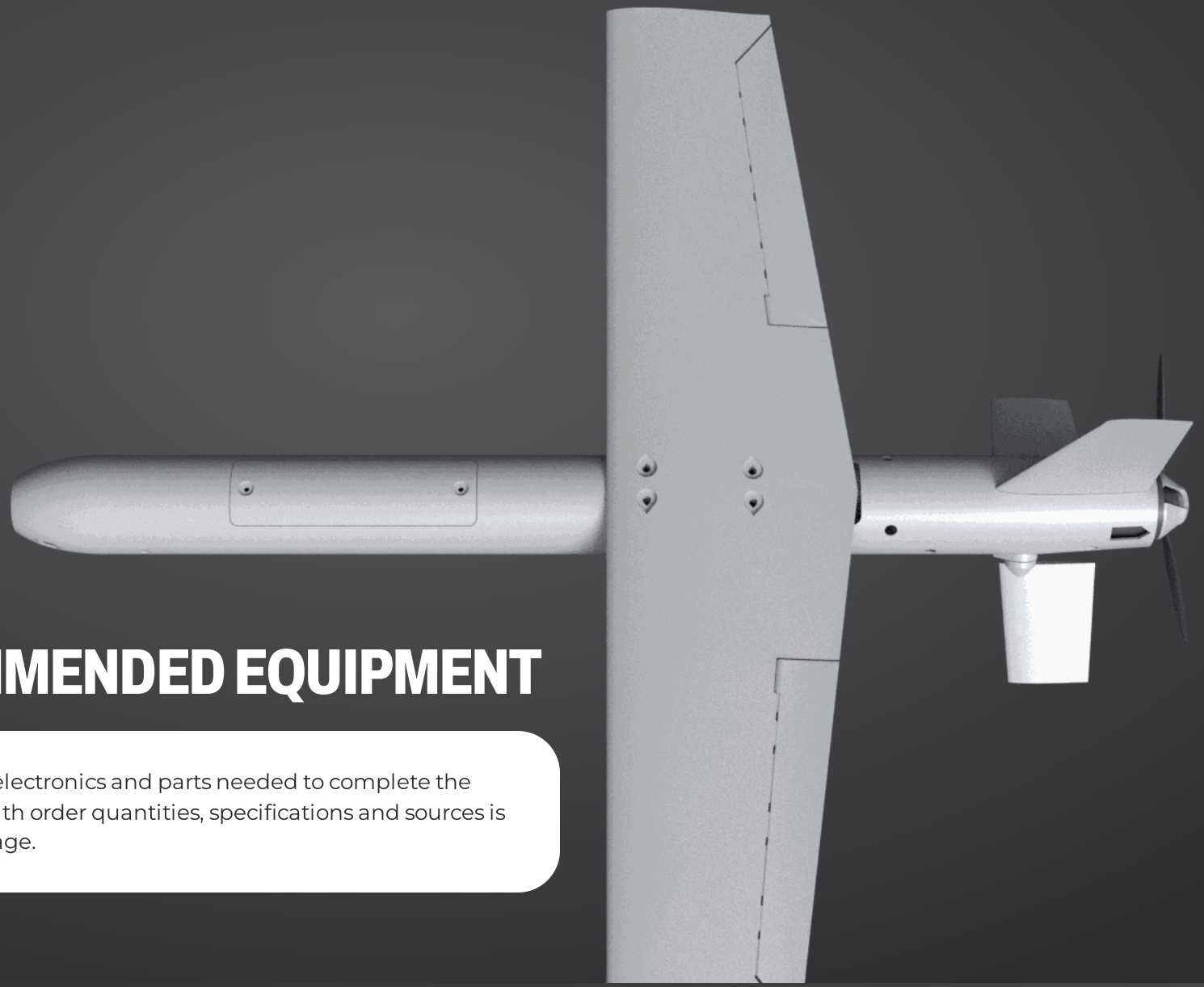
NOSE BAY
Camera and VTX (or the DJI O4 unit) in the removable nose



CAMERA NOSE VARIANTS



Nose	Camera	Battery centre for balance
N0M	19 mm micro FPV camera + 20 × 20 VTX	x 114 mm
N0P	Raspberry Pi Camera Module 3 (test option)	x 89 mm (at its stop + nose ballast)
N0MD	micro camera behind a 40 mm dome, nano VTX under the hatch	x 0 mm
N0PD	Pi camera behind a dome	x 0 mm
N0O	DJI O4 Air Unit (Lite), ducted	x 0 mm
N0OD	DJI O4 behind a dome	x 0 mm



RECOMMENDED EQUIPMENT

This section lists the electronics and parts needed to complete the aircraft. The full list with order quantities, specifications and sources is BOM.csv in the package.

RECOMMENDED ELECTRONICS

COMPONENT	MODEL / SPECIFICATION
Motor	2207 2550KV 4S racing motor
Propeller (fixed)	APC 5x7.5E (LP05075E)
ESC	HDZero Halo 4in1 70A ESC AM32
Flight controller	SpeedyBee F405 WING MINI
GPS + compass	TBS M10Q GPS GLONASS (QMC5883 compass), 13.5 g
Servos	3 × Emax ES9052MD or ES9052MD HV (Promodelhobby, Turkey)
Flight battery	Tattu R-Line 4S 650 75C (60 x 31 x 21 mm, 74 g)
Receiver	RadioMaster RP1 V2
FPV camera	Micro 19 mm FPV camera, e.g. RunCam Phoenix 2 or Caddx Ratel 2
Video transmitter + antenna	20x20 analog or digital VTX with U.FL antenna
DJI O4 Air Unit (Lite)	DJI O4 Air Unit

RECOMMENDED ACCESSORIES

ITEM	QTY
XT30 connector pairs	1
Carbon tube	1
Music wire 2.0 mm	2
Pushrods 1.2 mm with Z-bend	4
Pushrod keepers for 1.2 mm wire	4
Z-bend pliers, 1.1 mm drill, 1.2 mm drill (as a hand reamer), small flat file	0
Steel countersunk screw M3 x 16	2
M3 hex nut	4
M2 x 6 countersunk screws (joints)	4
Kapton / thin clear tape	as needed
Adhesives	as needed
Velcro strap + hook/loop	1
Heat shrink + silicone wire	as needed
Silver steel rod 2.5 mm	2
Brass tube 3 mm	2
MR30 connector pairs	1
Servo extension leads 100 mm	2

ITEM	QTY
Motor screws M3	4
Steel countersunk screw M3 x 12	2
Steel cup washer M3 (countersunk finishing washer)	4
Steel cup washer M2 (countersunk finishing washer)	2
M2 x 6 countersunk screws (motor plate)	2
Servo strap / zip tie + foam tape	1
Steel rod 1.5 mm	2
Vent tape	1
15T servo arms	1
Dupont housings + crimps (1-pin, 4-pin)	3
Carbon-fibre tube 40 x 38 mm	320 mm
M2 x 6 countersunk screws (tube)	8
Structural epoxy	~3 g
Thin CA	~1 g
M2 x 6 countersunk screws (hatch)	2
Conformal coating	~2 ml

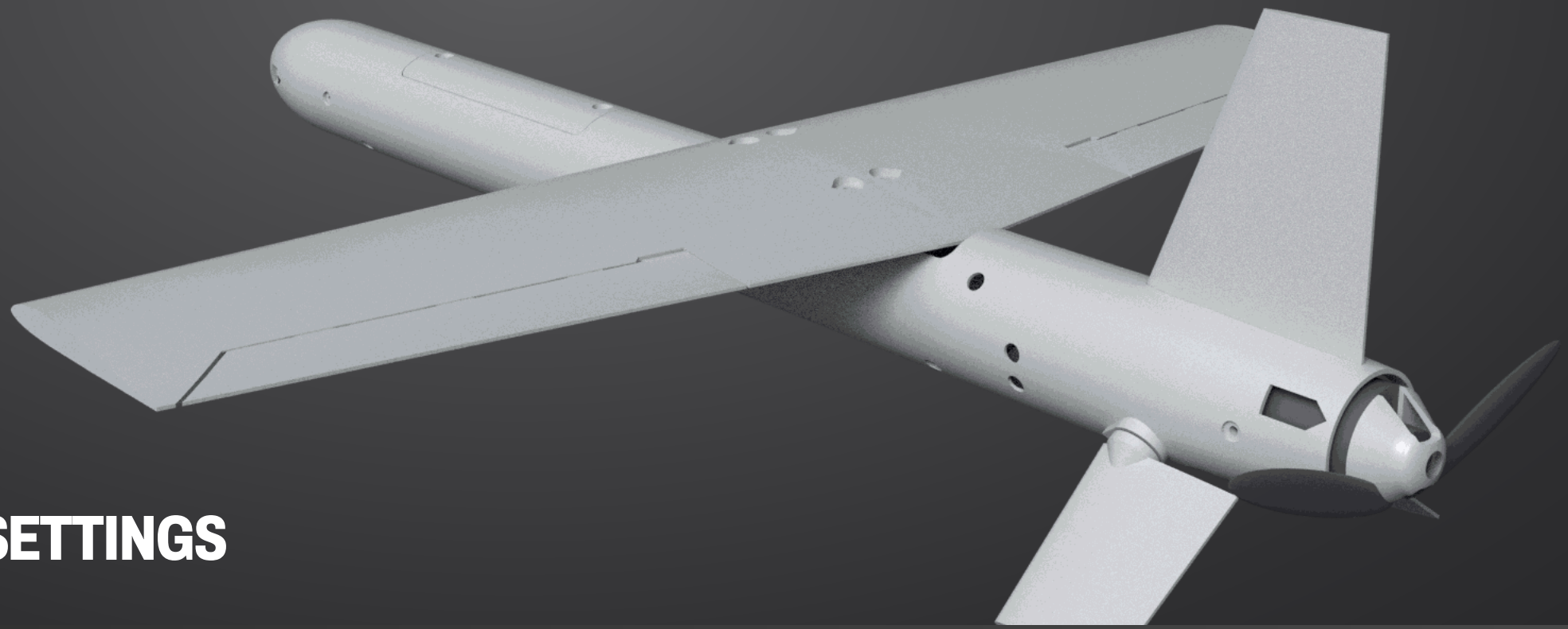
POWERTRAIN

Motor and prop. 2207 2550 KV class 4S racing motor (e.g. 2207 2550KV: 27.9-28 mm bell, about 21.5 mm body, 33.5 mm with the 5 mm prop shaft, 33 g, 16 x 16 mm M3). Prop: APC 5 x 7.5E fixed thin-electric pusher (LP05075E, 5 mm bore, hub 12.7 x 7.1 mm, 30,000 rpm rating), rpm-governed at 29,500.

Throttle cap. A 5-inch prop is rated about 30,000 rpm. On a full 4S pack the unloaded prop would exceed it at full throttle, so the flight controller caps the throttle at 70 %. Never raise it, and inspect the blades after every thrust-stand run and flying day.

rpm governor (open item). This version is designed around a governor that holds the prop at 29,500 rpm, which lets it use the full rated rpm in level flight. ArduPilot's governor script does not run on the F405 flight controller, so until an H7 flight controller or a governing ESC is fitted the throttle cap above applies and the top speed is about 63 m/s; the 80–90 m/s stages need the governor.

Battery. Tattu R-Line 4S 650 75C (60 x 31 x 21 mm, 74 g); XT30. Full throttle in bursts of ≤ 10 s, then ≥ 30 s at low throttle: the ESC lives on its thermal mass. Static thrust in the model: 1064 g at 51 A; confirm on a thrust stand before the first flight.



PRINT SETTINGS

PRINT SETTINGS

Filament per part (no glass skin): **PETG-CF** wing halves (2 walls), ailerons, equipment tray, tail coupler, nut bars, hatch lips, motor plate and spinner; **PETG** camera noses, hatch cover and GPS carrier (radio windows: never carbon-filled filament there); **ASA** tail module and fin (motor heat; the fin carries the receiver antenna) and the DJI duct insert (next to the hot O4 module); PETG-CF tail panels and servo horns. 0.4 mm hardened-steel nozzle, 0.20 mm layers; every part fits both the H2S and the H2D build volume (340 × 320 × 325 mm).

GENERAL GUIDELINES

- Fuselage, nose and hatch shells: 2 walls, 30 % lightning infill (it only grows under top surfaces: at 0 % the solid bosses and bridges print hollow and their tops collapse).
- Wing halves: PETG-CF, 2 walls, 8 % gyroid. Ailerons: PETG-CF, 1 wall, 5 % gyroid.
- Tail panels: 2 walls, 10 % gyroid. Fin: 2 walls, 15 %.
- Filament per part: see the parts list (PETG-CF, PETG, ASA). CF filaments need a hardened-steel nozzle; ASA an enclosed printer.
- Tall standing parts: 5 mm brim, slow outer wall, enclosure door closed.

SUPPORTS: OFF

Every part is designed to print without supports in the orientation shown, and checked on the slicer's own G-code with these settings: nothing is printed over air and no bridge is longer than 10 mm. Keep supports switched off and use the supplied .3mf files (they carry the settings). The downloads hold ONE set of Bambu Studio projects for the H2S and the H2D: no G-code inside (G-code only fits the printer it was sliced for), tree supports on (auto). Open, select your printer, slice, print. Check the top of the aileron's torque-rod bore on the first aileron.

PRINT SETTINGS

FIRST: MEASURE THE TUBE

Measure the carbon tube's bore at both ends before printing anything that fits inside it (the design uses 38.0 mm). If it differs by more than 0.1 mm, the inside parts must be rebuilt for your tube. Print the tail coupler first and try it in both ends of the tube: it must slide in with a light push.

COUPON TESTS

Print the camera nose and try its spigot in the tube's front; slide the tail module onto the coupler; print the tray and slide it the whole length of the tube before cutting any window. Print a 40 mm wing-root section in PETG-CF and check the spar fit.

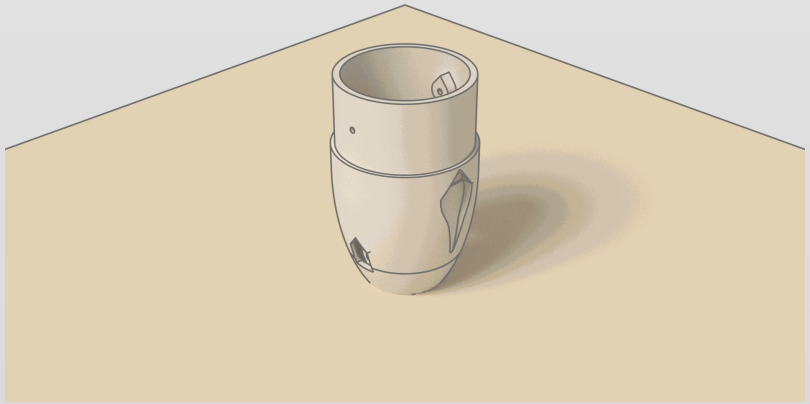
PARTS LIST

PART	MATERIAL	MASS
N0M NOSE (MICRO FPV)	PETG	13.9 g
N0P NOSE (PI CAMERA)	PETG	13.2 g
N0O NOSE (DJI O4)	PETG	14.7 g
F3 TAIL MODULE	ASA	14.3 g
HF FRONT HATCH	PETG	4.3 g
FN FIN	ASA	6.7 g
VTX SHELF (N0M)	PETG	0.8 g
HG GPS CARRIER	PETG	1.1 g
NB1 WING-NUT BAR FRONT	PETG-CF	1.2 g
NB2 WING-NUT BAR REAR	PETG-CF	1.3 g
HL1 HATCH LIP FRONT	PETG-CF	0.9 g
HL2 HATCH LIP REAR	PETG-CF	0.9 g

PART	MATERIAL	MASS
WR WING HALF RIGHT	PETG-CF	46.9 g
WL WING HALF LEFT	PETG-CF	47.0 g
AR AILERON RIGHT	PETG-CF	4.0 g
AL AILERON LEFT	PETG-CF	3.8 g
SR TAIL PANEL RIGHT	PETG-CF	5.2 g
SL TAIL PANEL LEFT	PETG-CF	5.2 g
M1 MOTOR PLATE	PETG-CF	2.1 g
SP SPINNER	PETG-CF	1.6 g
TR EQUIPMENT TRAY	PETG-CF	17.8 g
TC TAIL COUPLER	PETG-CF	8.5 g
CW COUNTERSUNK WASHER (PRINT 14)	PETG-CF	0.1 g
WF WING-WINDOW FILLER	PETG-CF	1.4 g

One aircraft with the N0M nose: 238 g printed. Only one camera nose is needed; the others are options.

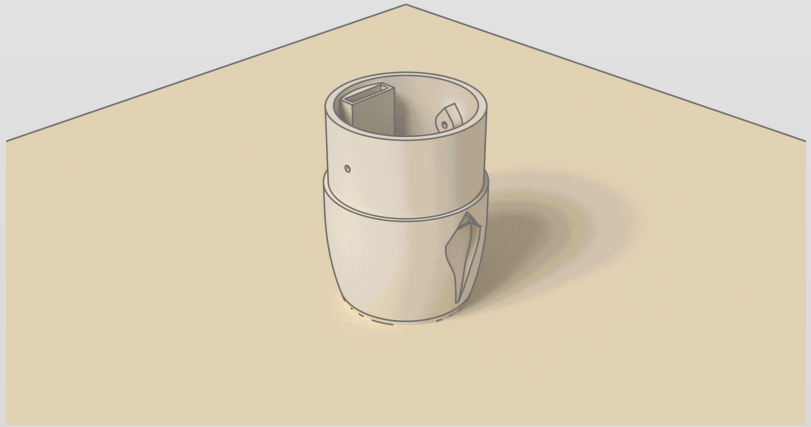
PARTS ORIENTATION



N0M NOSE (MICRO FPV)

FACE DOWN, SPIGOT UP

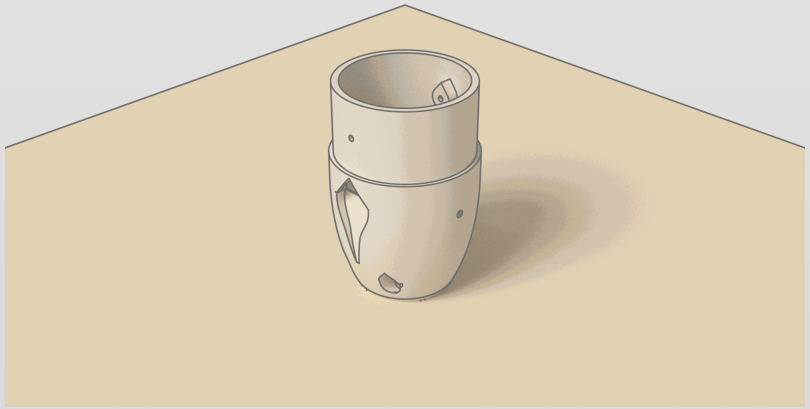
2 WALLS · 30 % INFILL · PETG



N0P NOSE (PI CAMERA)

FACE DOWN, SPIGOT UP

2 WALLS · 30 % INFILL · PETG

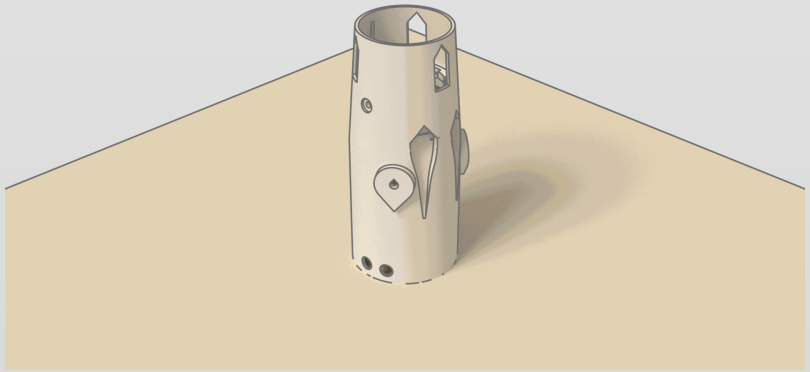


N0O NOSE (DJI O4)

FACE DOWN

2 WALLS · 30 % INFILL · PETG

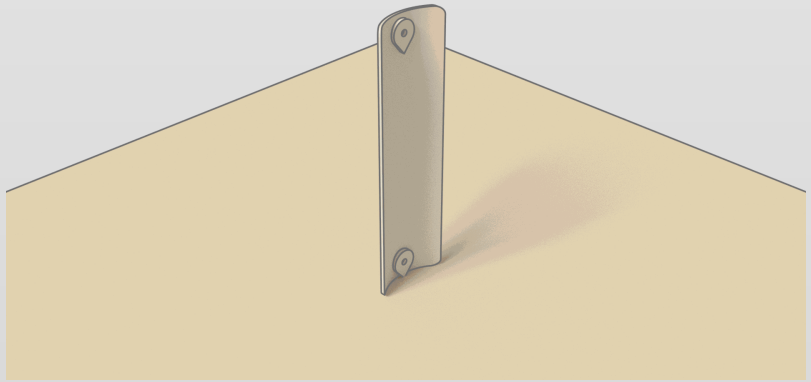
PARTS ORIENTATION



F3 TAIL MODULE

FRONT END (SOCKET) DOWN · 5 mm BRIM

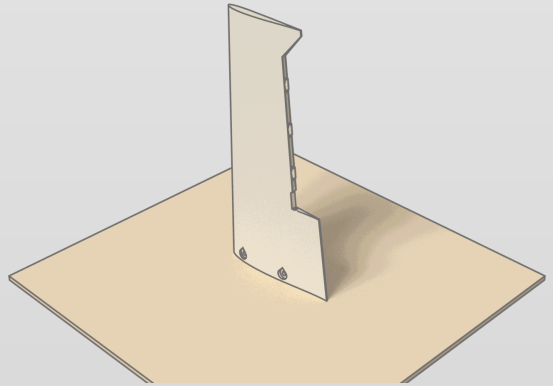
2 WALLS · 30 % INFILL · ASA



HF FRONT HATCH

STANDING, FRONT END DOWN

2 WALLS · 30 % INFILL · PETG

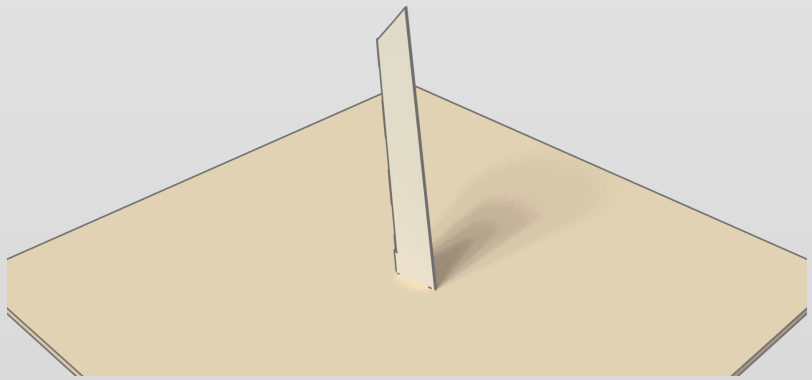


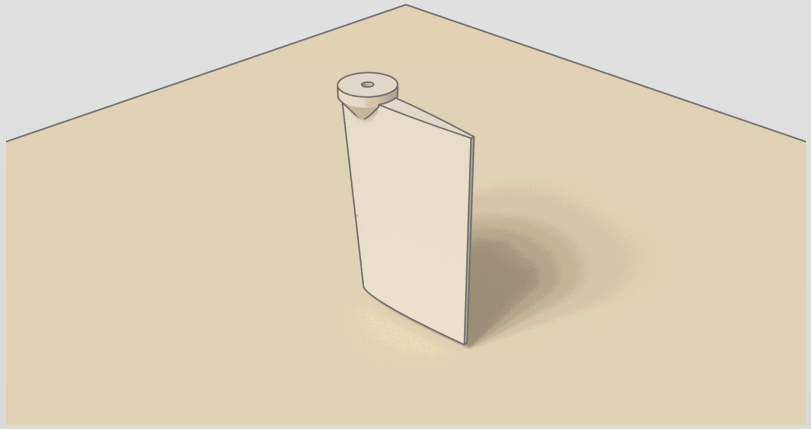
WR WING HALF R / L

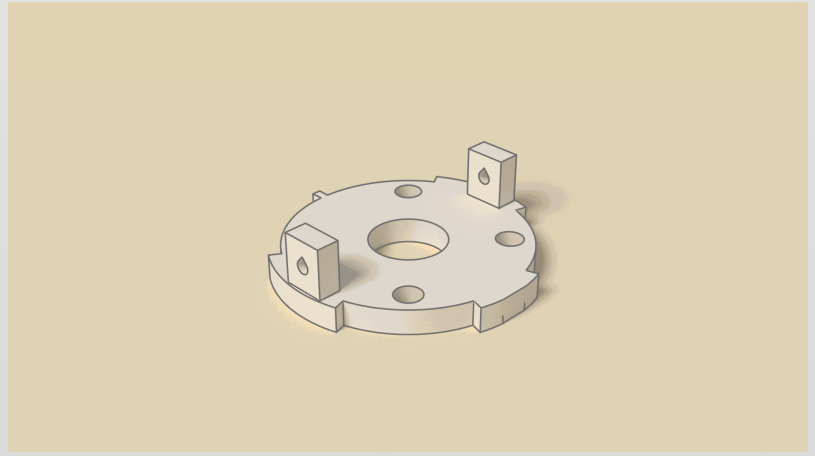
ON THE ROOT RIB

2 WALLS · 8 % INFILL · PETG-CF

PARTS ORIENTATION







AR AILERON R / L

ON THE INBOARD END

1 WALLS · 5 % INFILL · PETG-CF

SR TAIL PANEL R / L

TIP DOWN, HUB UP

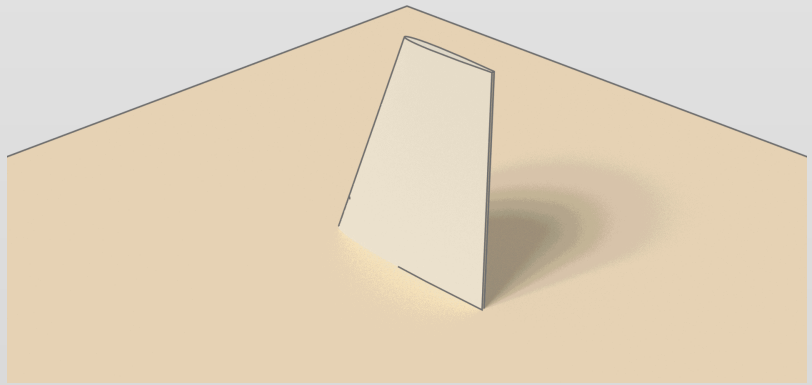
2 WALLS · 10 % INFILL · PETG-CF

M1 MOTOR PLATE

FLAT

4 WALLS · 40 % INFILL · PETG-CF

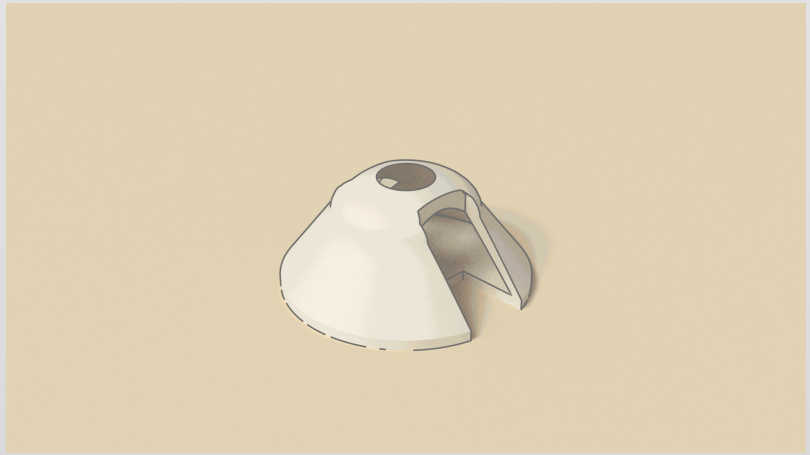
PARTS ORIENTATION



FN FIN

ON THE ROOT TAB

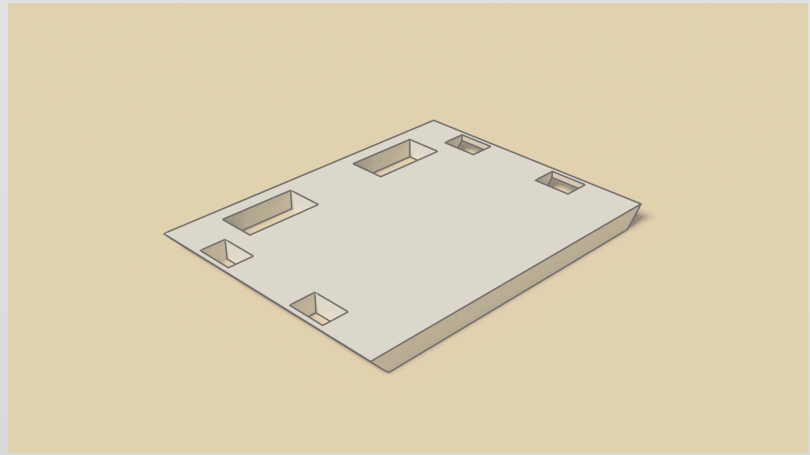
2 WALLS · 30 % INFILL · ASA



SP SPINNER

BACKPLATE DOWN

4 WALLS · 40 % INFILL · PETG-CF

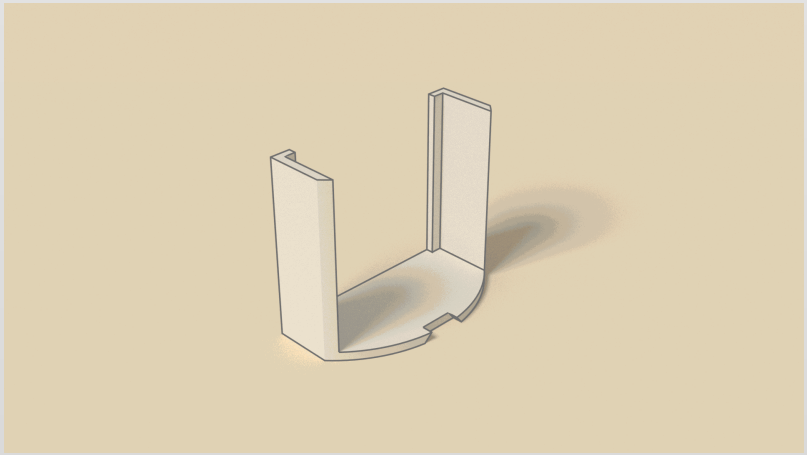


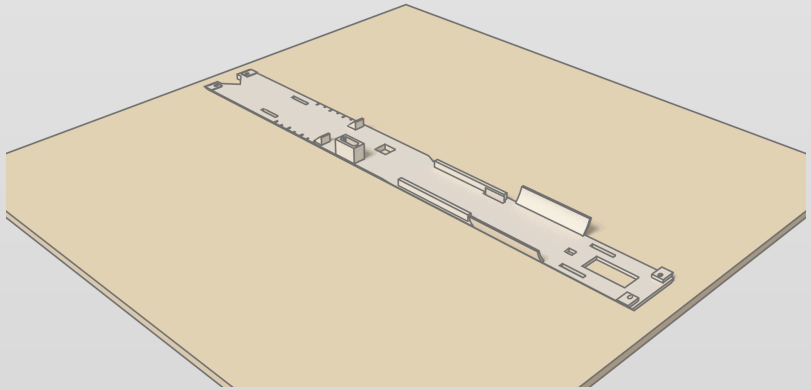
VTX SHELF (N0M)

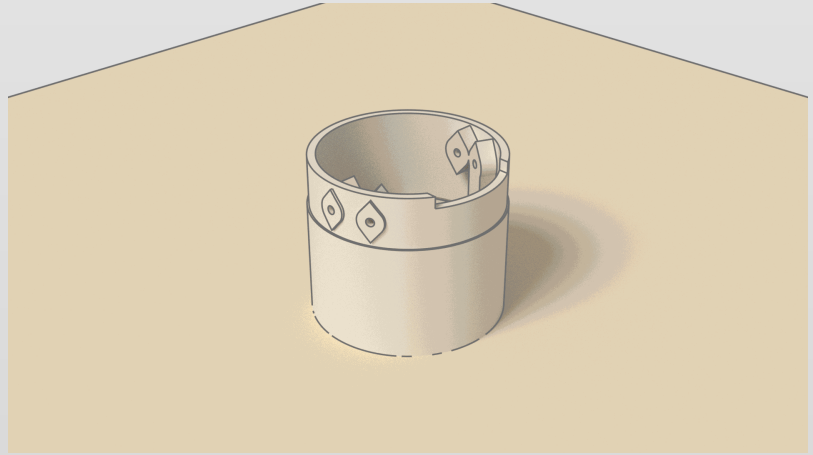
FLAT

2 WALLS · 30 % INFILL · PETG

PARTS ORIENTATION







HG GPS CARRIER

ON ITS REAR WALL

2 WALLS · 30 % INFILL · PETG

TR EQUIPMENT TRAY

UPRIGHT ON ITS FLAT UNDERSIDE

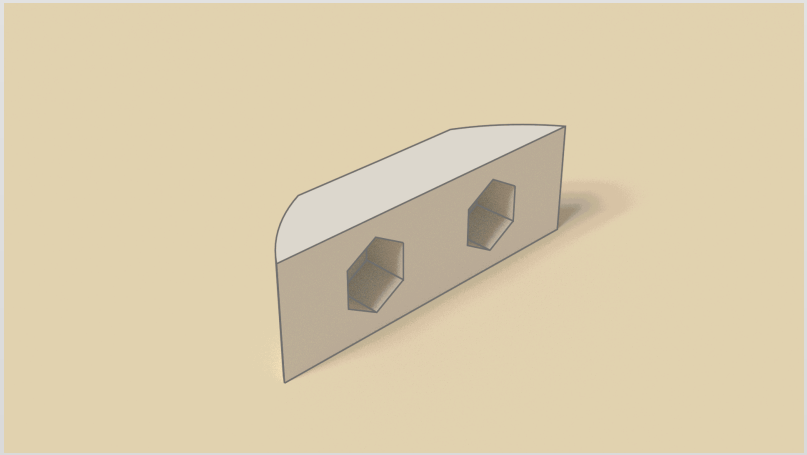
2 WALLS · 30 % INFILL · PETG-CF

TC TAIL COUPLER

ON ITS SPIGOT END

2 WALLS · 30 % INFILL · PETG-CF

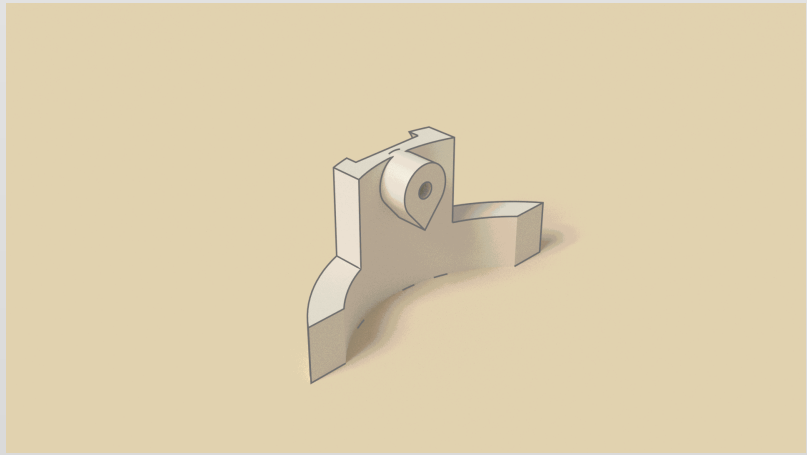
PARTS ORIENTATION



NB1 WING-NUT BAR FRONT

ON ITS END FACE

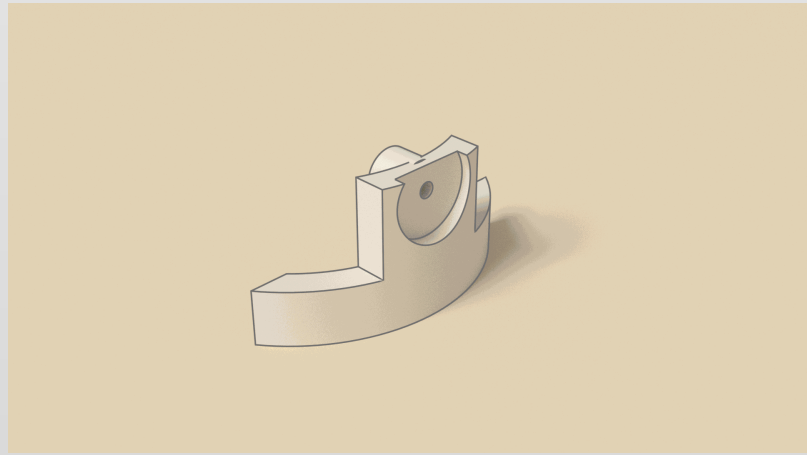
2 WALLS · 30 % INFILL · PETG-CF



HL1 HATCH LIP FRONT

BONDING FACE DOWN

2 WALLS · 30 % INFILL · PETG-CF

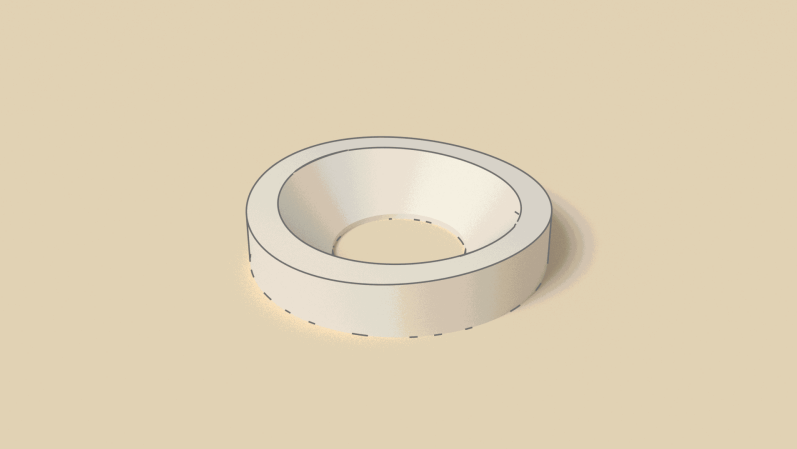


HL2 HATCH LIP REAR

BONDING FACE DOWN

2 WALLS · 30 % INFILL · PETG-CF

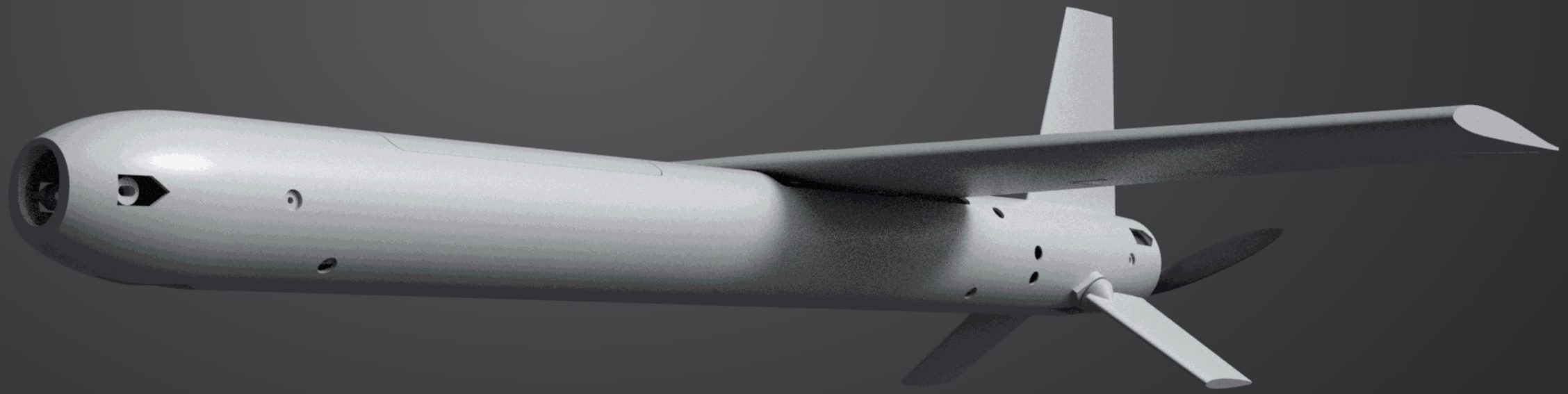
PARTS ORIENTATION



CW COUNTERSUNK WASHER (PRINT 14)

FLAT FACE DOWN

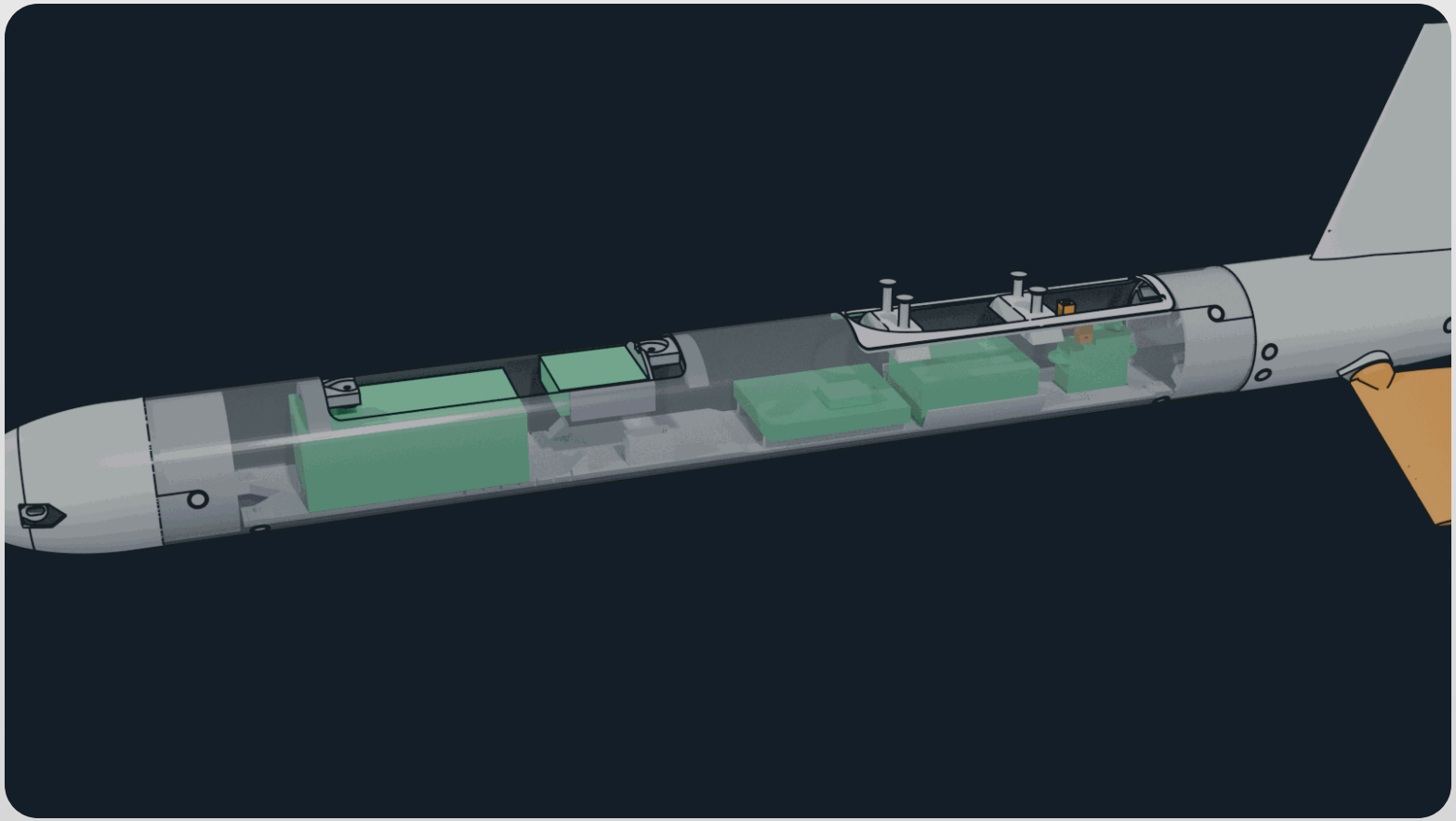
2 WALLS · 30 % INFILL · PETG-CF



ASSEMBLY GUIDE

CARBON TUBE

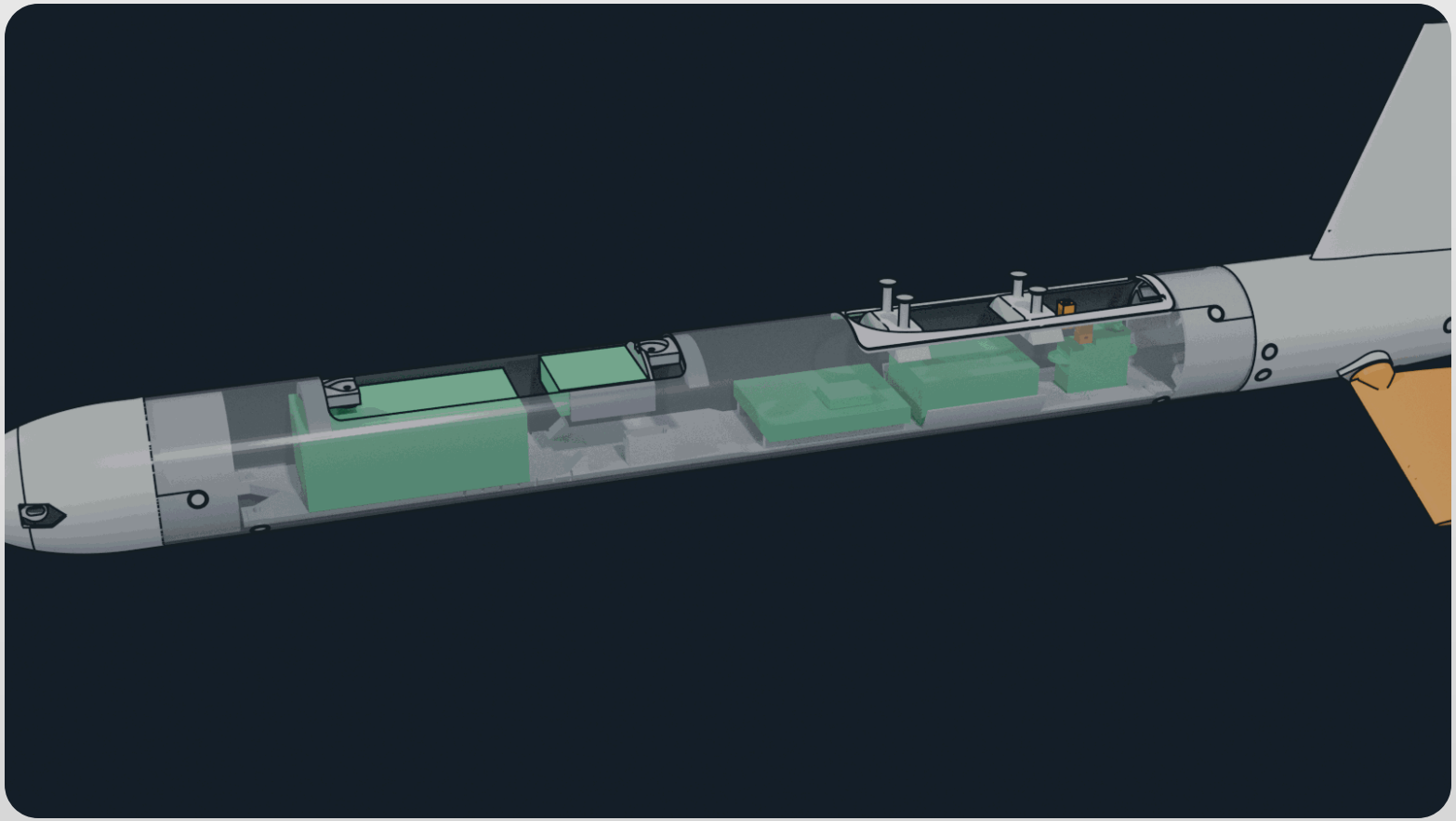
Cut the 40 × 38 mm carbon tube to 320 mm. Print the 1:1 cutting template (two A4 pages, 100 % scale: check its 100 mm bar), cut the strip out, wrap it round the tube with its ends on the tube ends and the TOP line along the top, tape it. Cut every window inside the red lines with a diamond disc and file to the line; drill the 2.2 mm holes. FFP3 mask, eye protection, vacuum at the disc or cut wet. Seal every cut edge and hole with thin CA: carbon conducts, a bare edge can cut a wire's insulation and short it.



DIAMOND DISC

CARBON TUBE

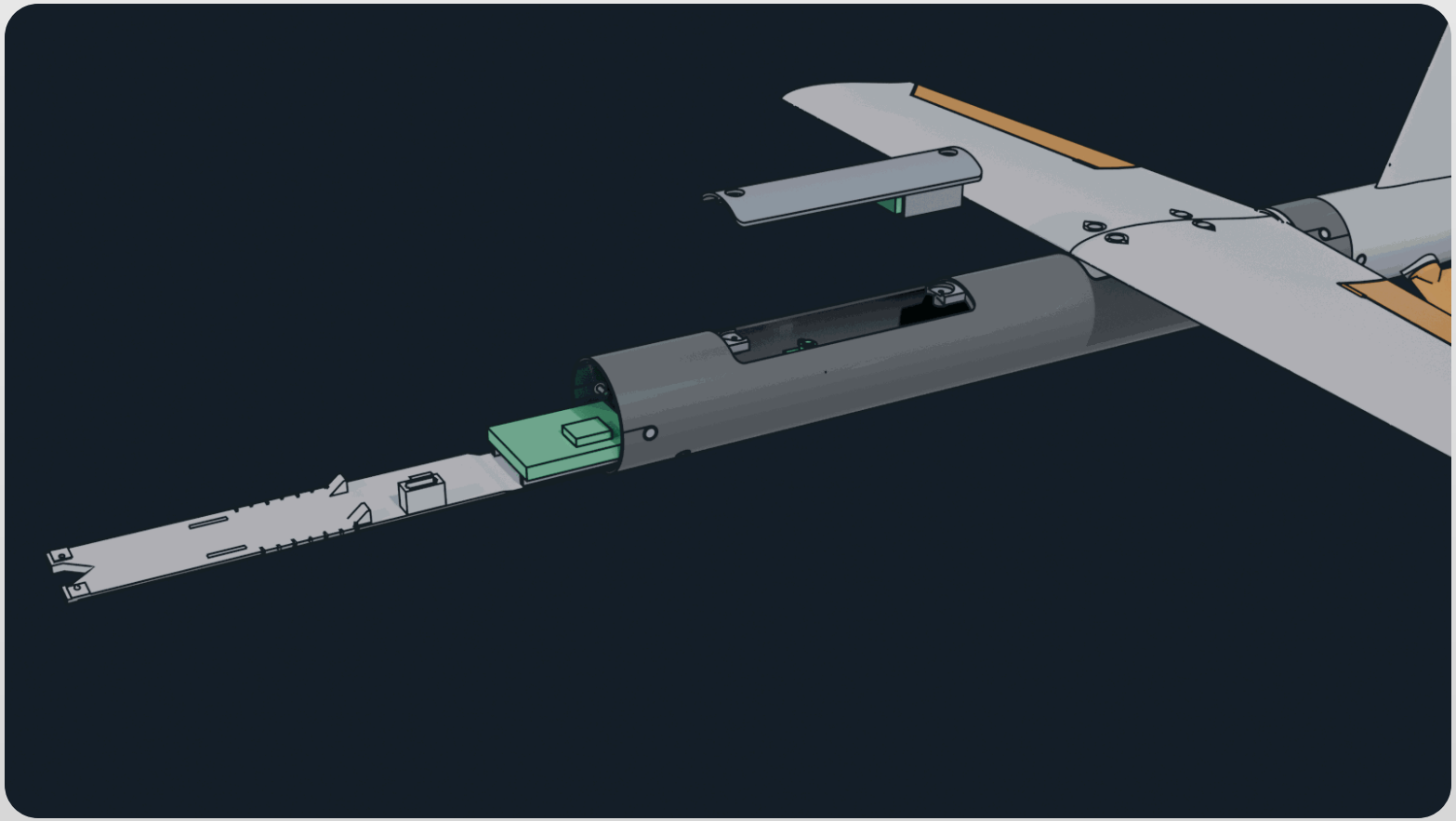
Scuff the bonding areas inside the tube (120 grit) and clean them with alcohol. With structural epoxy bond: the tail coupler TC into the aft end, its sleeve over the whole ring behind the wing window, and fix it with 2 × countersunk M2 × 6 in washers at the sides (they hold it aligned while the epoxy cures and back the bond up); the two wing-nut bars NB1/NB2 across the wing-bay window (in through the window, turned under its edges, pocket down); the hatch lips HL1/HL2 under the front hatch opening's ends (their screw bosses down, centred under the opening). Let it cure before the tray goes in.



EPOXY

EQUIPMENT TRAY

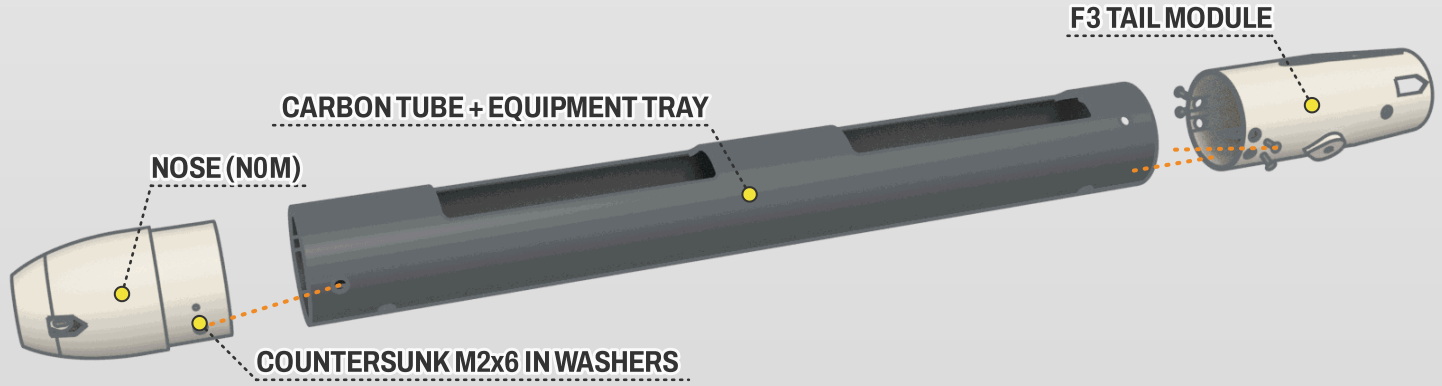
Build the tray on the bench: the ESC on its rails, the FC stack, receiver and aileron servo, all wiring as on the next pages. Take the camera nose off, push the tray into the tube from the front until its screw holes line up, and fix it with 4 × countersunk M2 × 6 screws through the tube (do not overtighten: they thread into printed pillars). Every external screw (the tube, the tail joint, the motor plate) sits flush in a printed countersunk washer (CW) pressed into its 5 mm hole: print 14, push one into each hole before the screw. To service the equipment: battery out, nose off, unhook the aileron pushrods, unplug the tail-module plugs, four screws out, slide the tray forward.



COUNTERSUNK M2 × 6

FUSELAGE ASSEMBLY

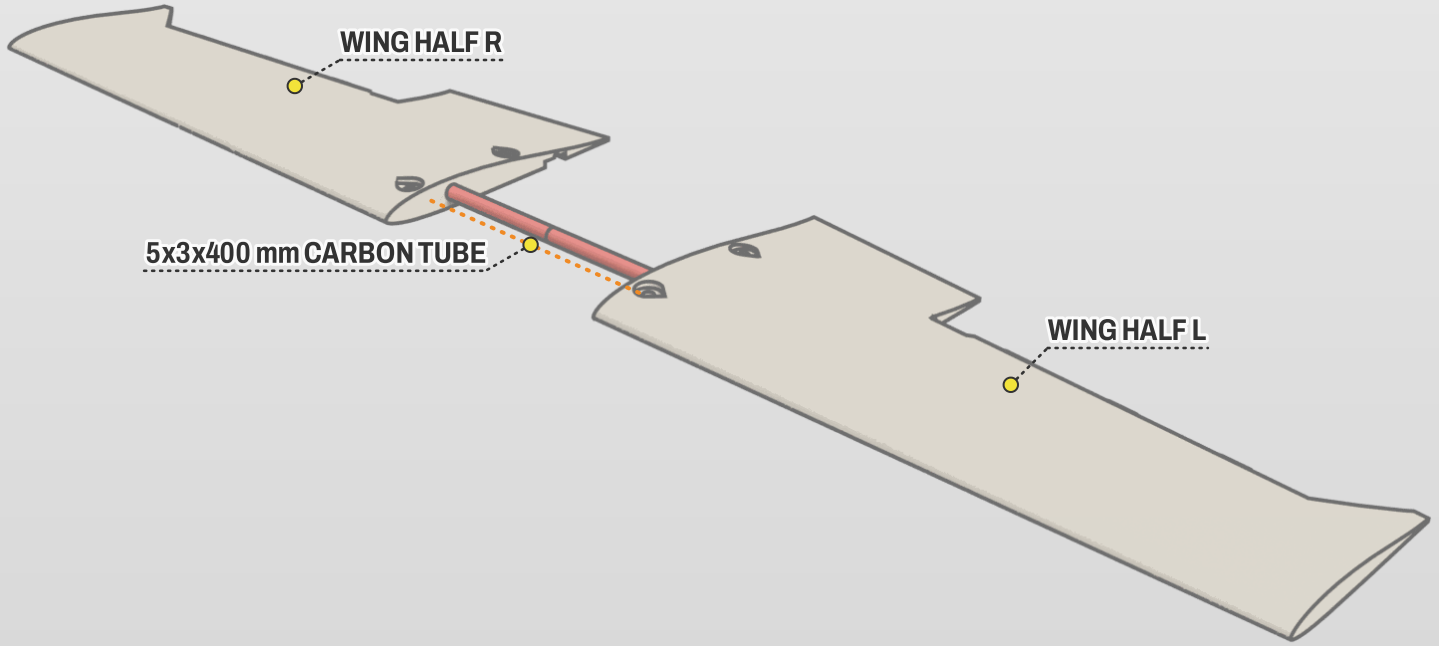
The camera nose plugs into the tube's front (its spigot is sized to the tube's bore) with 2 × countersunk M2 × 6, flush in their printed washers, into its bosses; the tail module slides onto the tail coupler's spigot (the four pads on the coupler slide in the grooves inside the module) and is held by 4 × countersunk M2 × 6 (two each side: at the side and 25° below it), each in a printed washer in its 5 mm hole, as on the tube. Nothing is glued: both modules come off for service. They must slide on by hand; if the bore is tight, sand the spigot, never force it.



M2 × 6 SCREWS

WING ASSEMBLY

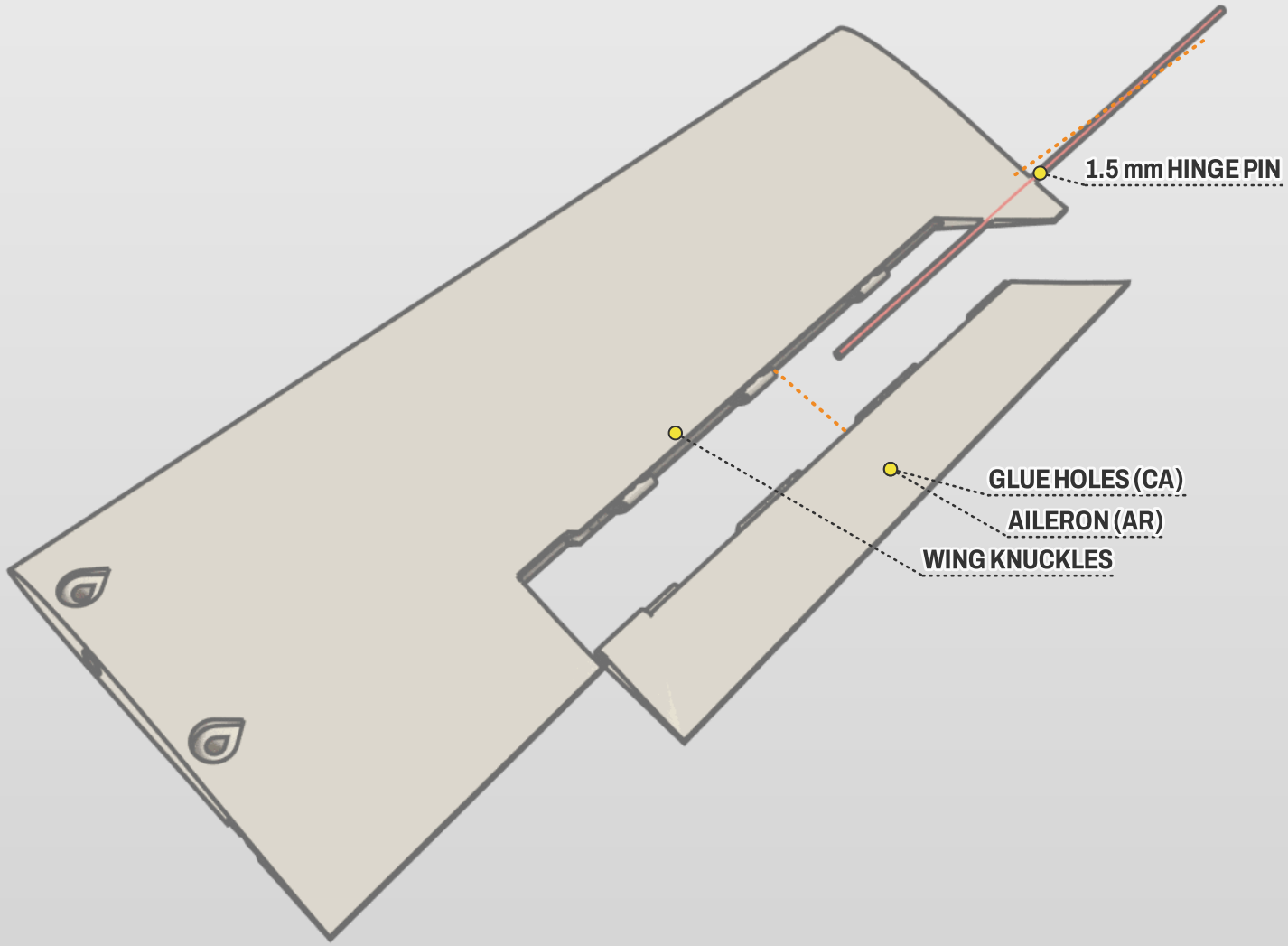
Cut one 400 mm length of 5 × 3 mm carbon tube and slide it through both wing halves (200 mm into each). Epoxy the tube in the bores and the two root ribs together with the whole wing flat on a table (0° dihedral).



EPOXY

WING ASSEMBLY

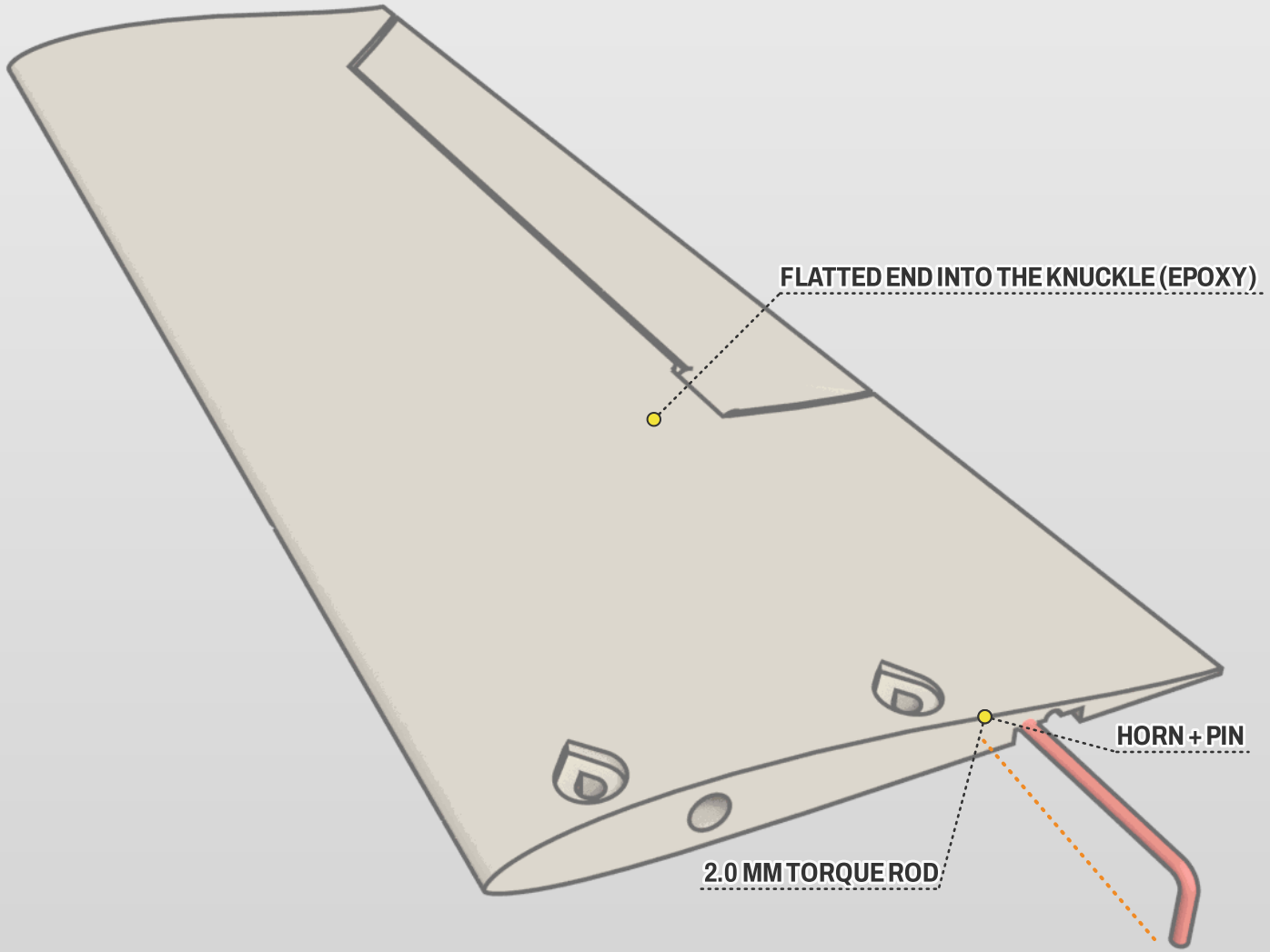
Set each aileron in place: its rounded nose sits in the cove along the wing's hinge face and the wing's three knuckles enter the pockets in the nose. Push a 132 mm length of 1.5 mm steel rod in from the wingtip through the tip, the aileron and the knuckles. Wick thin CA into the aileron's four glue holes only, then plug the tip hole. The aileron must swing freely.



THIN CA

WING ASSEMBLY

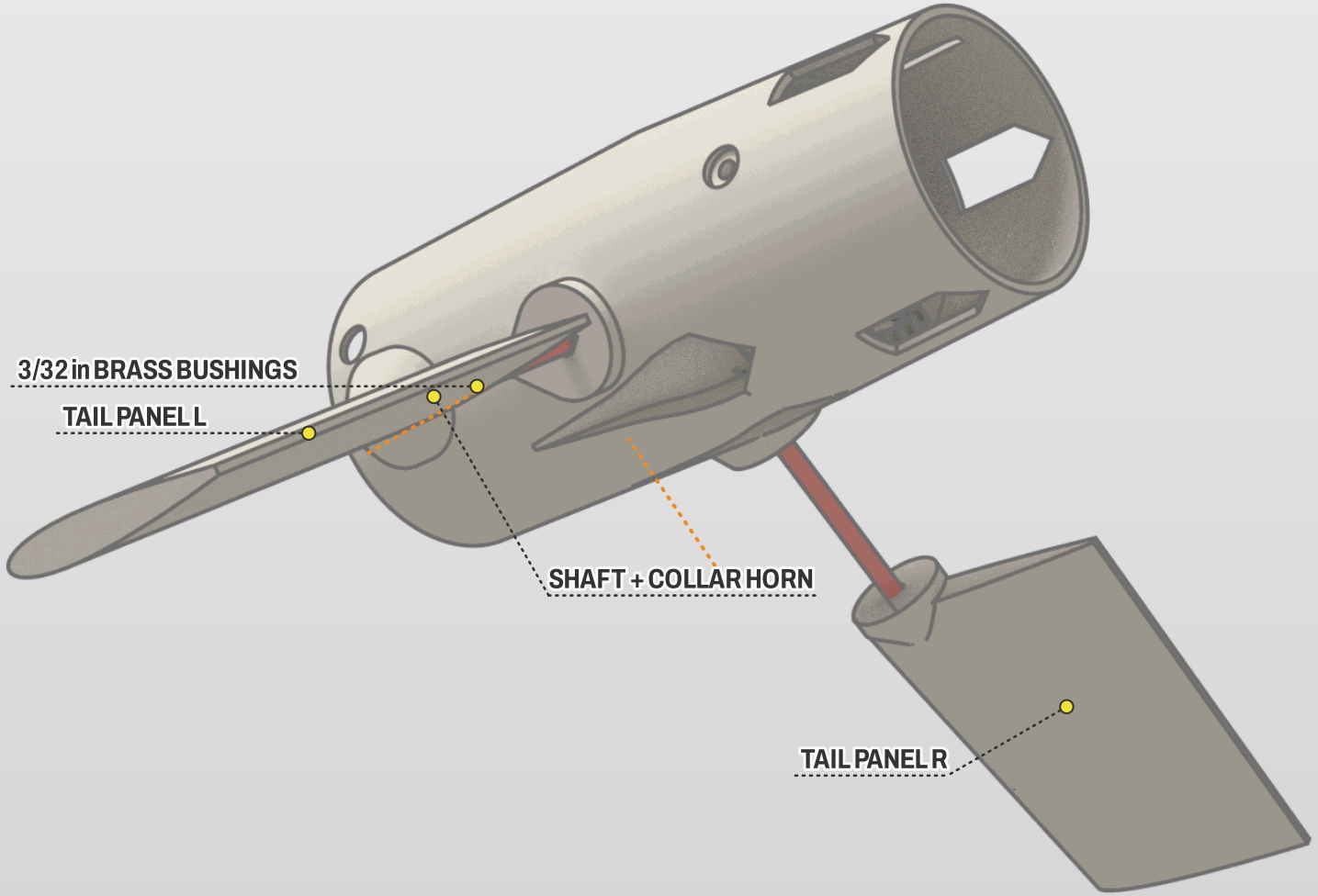
Bend a 12 mm horn down at one end of each 2.0 mm rod with a 5.5 mm pin outboard at its tip, and file a 0.3 mm flat on the last 9 mm of the straight end, opposite the horn. Push it in from the root face until the flat enters the D-bore of the aileron's root knuckle; epoxy the end and hold the aileron 3° trailing edge up while it cures.



EPOXY

TAIL MODULE

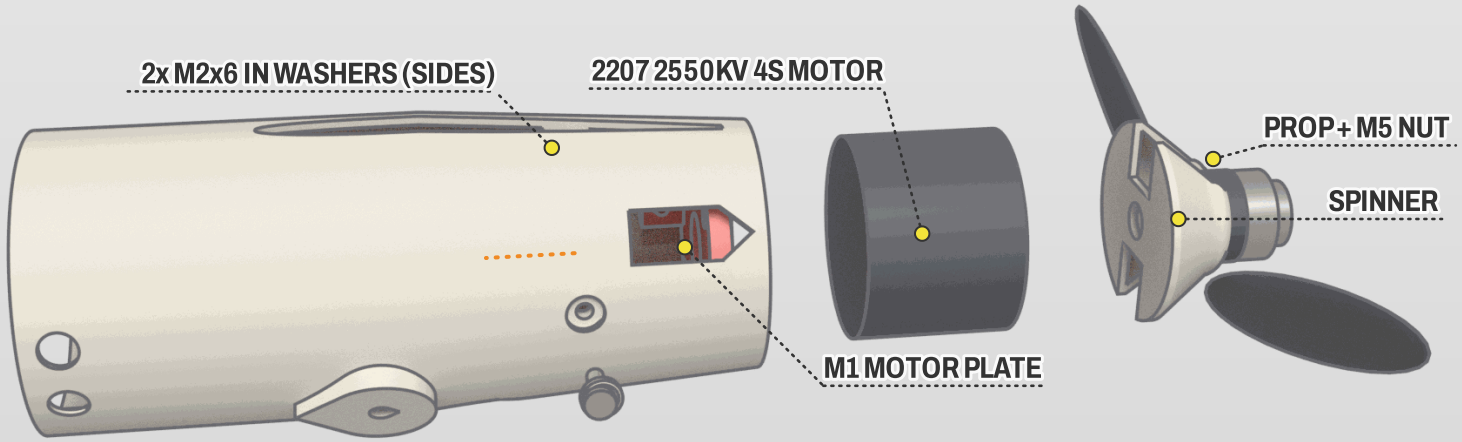
Press a 10 mm brass bushing into each side boss of F3. Epoxy a 2.5 mm steel shaft 35 mm into each tail panel's root bore. From the open front of F3, push each shaft through its bushing and fit the wheel-collar horn with its 8 mm brass plate standing along the panel's normal (not pointing forward), 0.5 mm end play, set screw on a filed flat.



EPOXY

TAIL MODULE

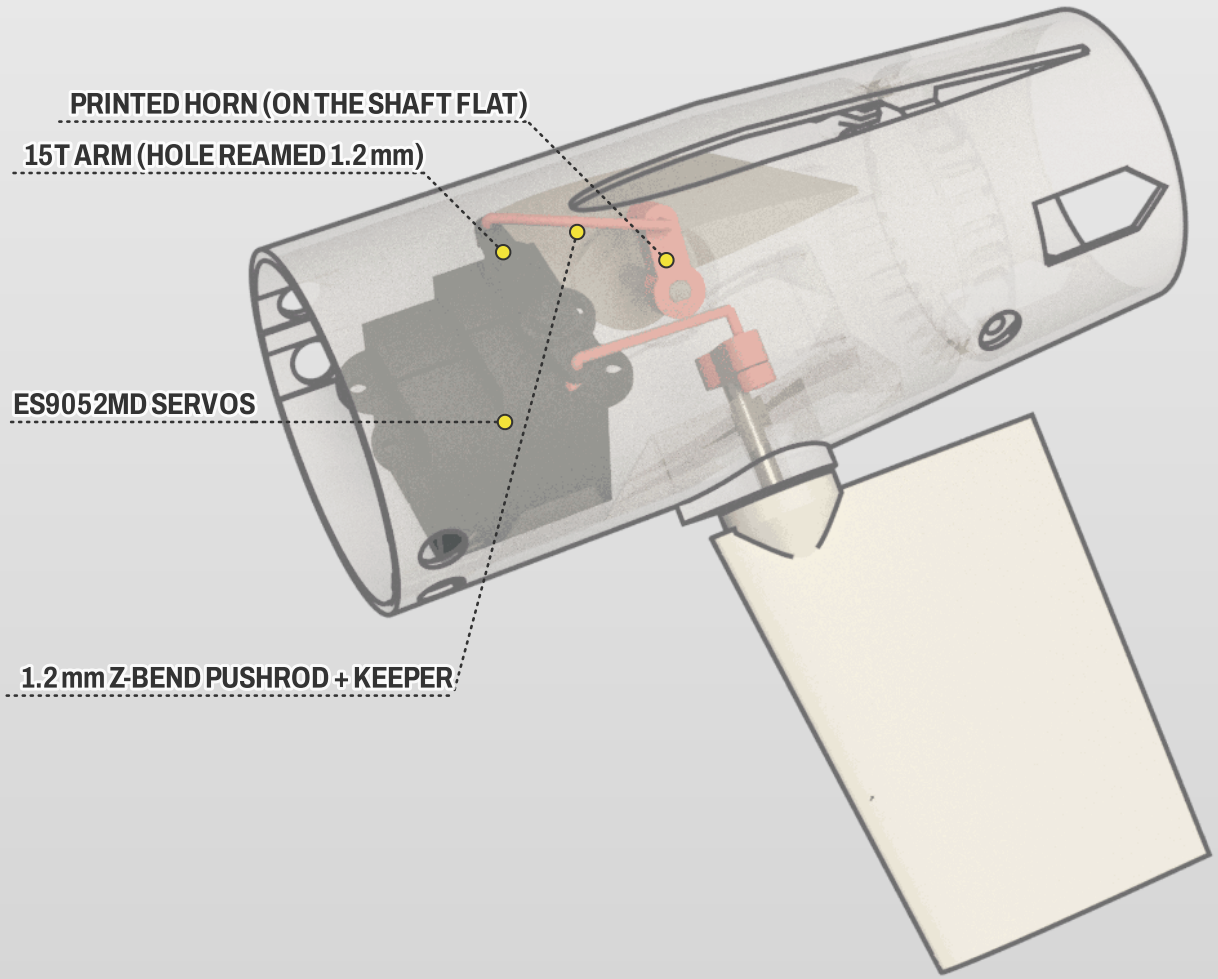
Screw the motor to the M1 plate on the bench with its 4 × M3 (heads on the plate's front face). Slide the plate into the rear of F3 until it sits on the seat ring, its two side lugs in the ring's pockets and its edge notches on the cooling windows, then fix it with 2 × countersunk M2 × 6 through the tail module's sides into the lugs, each in a printed washer in its counterbore, flush (no glue: the plate comes out for a different motor). Fit the spinner backplate, the prop (APC 5 x 7.5E fixed thin-electric pusher (LP05075E, 5 mm bore, hub 12.7 x 7.1 mm, 30,000 rpm rating), rpm-governed at 29,500) and the thin M5 nut.



M2 × 6 SCREWS

TAIL MODULE

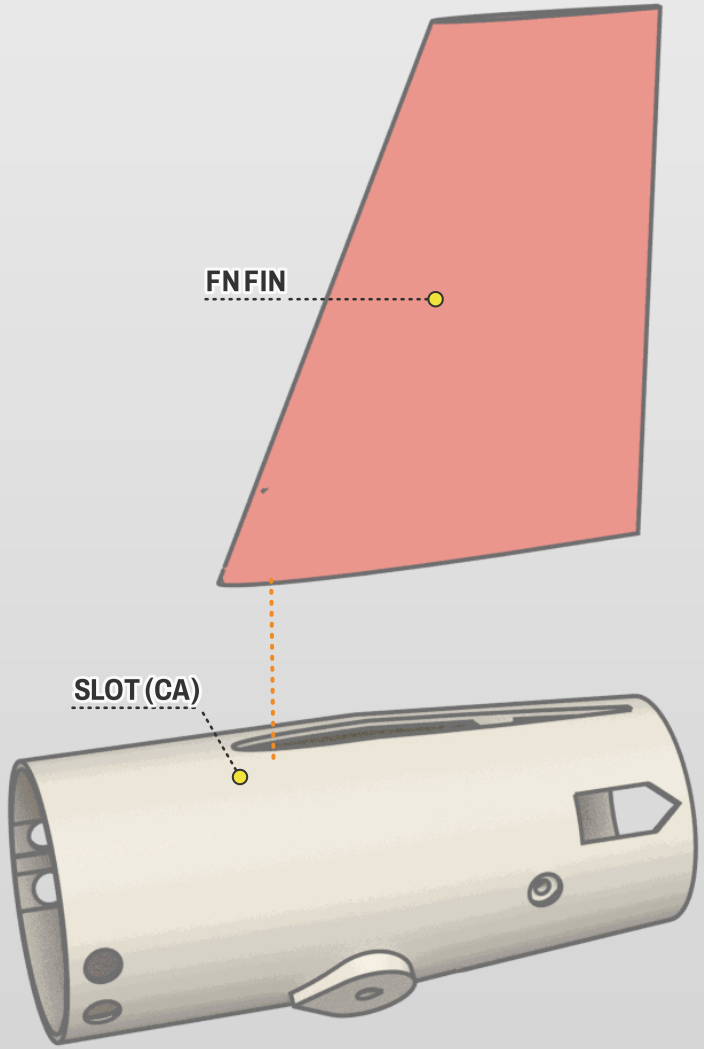
Stand both ES9052MD servos through the cut-outs in the F3 floor on foam tape on the bottom skin, a strap over them, output at the aft end, leads along the side walls. One-sided 15T arm pointing outboard; file a 0.3 mm flat on the inboard end of each 2.5 mm stabilator shaft and epoxy the printed horn (THR / THL) on it, arm along the panel's normal. Drill the servo arm's 6.7 mm hole 1.1 mm and ream it by hand with a 1.2 mm bit (and the horn's hole the same): the wire must turn with no play. 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend at the horn end pushed outboard through the horn's reamed hole, snap-on keeper on the protruding end (unclip it to take the tail module off). Free play at each panel trailing edge <= 0.2 mm. Check both panels move freely through ±20°.



SERVO TAPE

TAIL MODULE

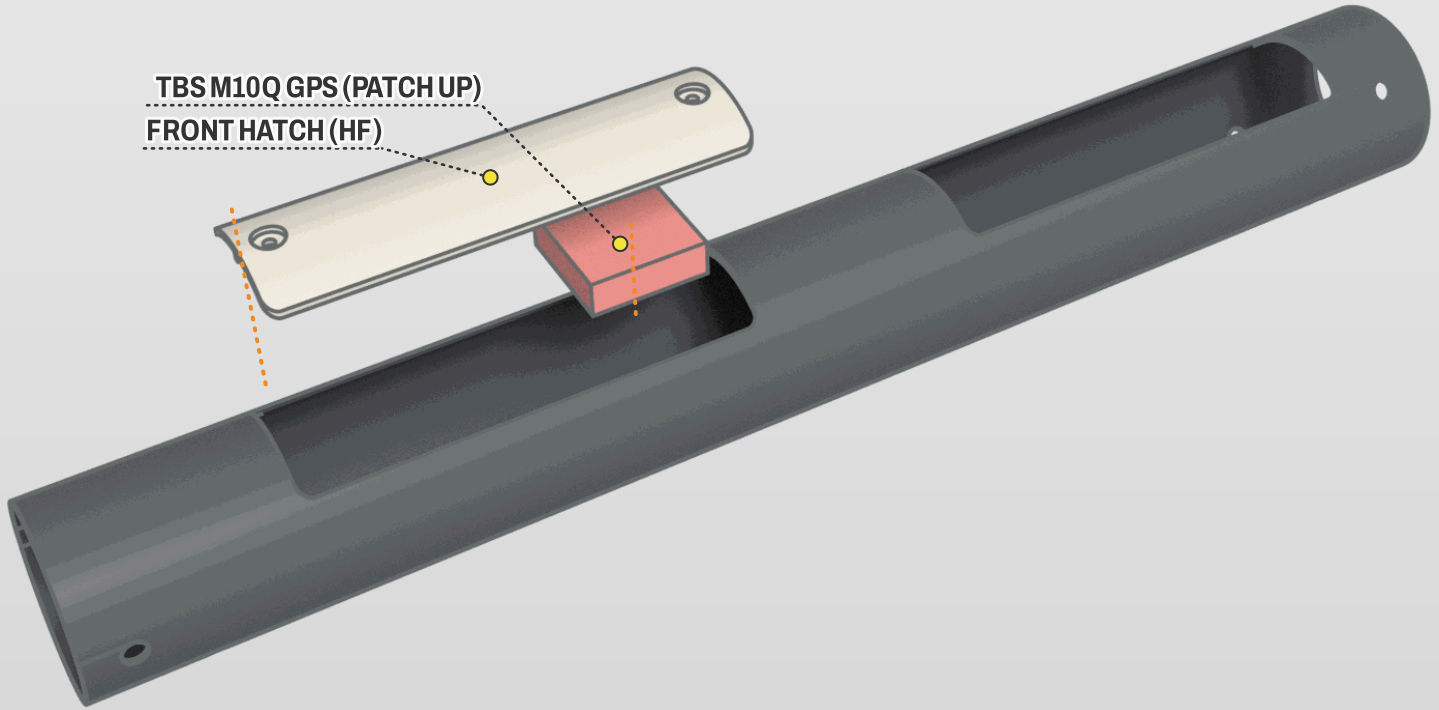
Glue the fin into the top slot of F3 with CA, vertical and square to the wing.



CA GLUE

FUSELAGE ASSEMBLY

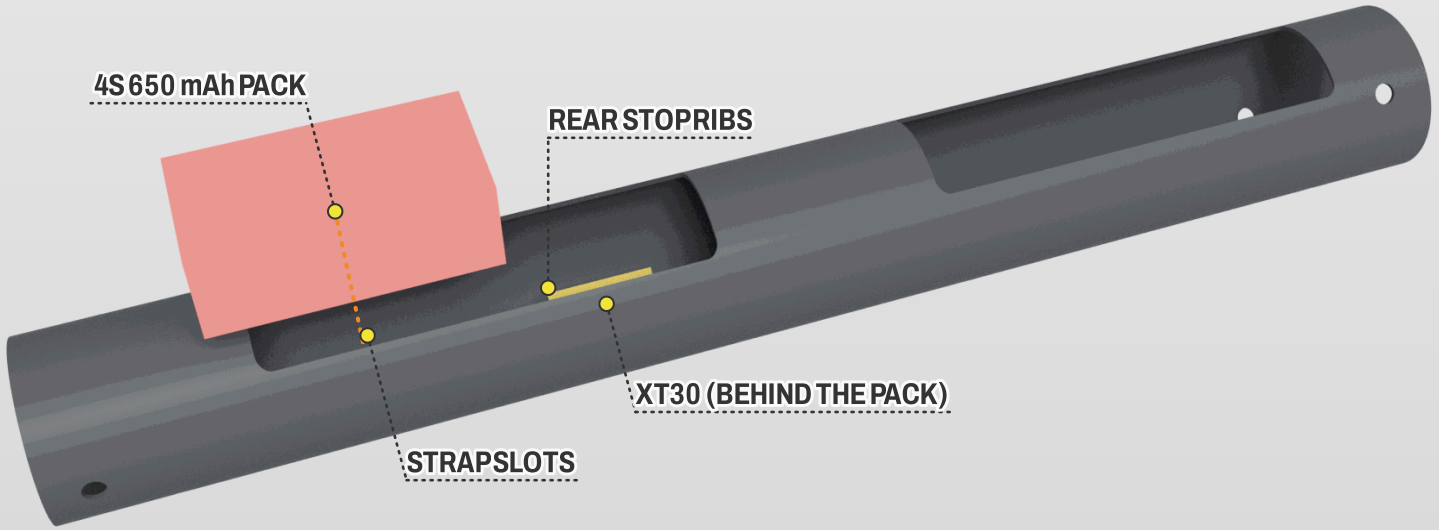
Bond the printed GPS carrier (HG) under the rear of the PETG hatch cover (the GPS's radio window: never paint it with metallic paint). Put foam tape on the carrier's two side lips and slide the TBS M10Q GPS in from the front, patch up, onto the lips against the rear wall: it is carried from below, with nothing but air and the cover between its patch and the sky. The cover is held by two flush countersunk M2 × 6 screws in steel cup washers, one at each end, into the bosses under the bonded lips (no magnets: suction and the GPS's weight would lift them at speed). Leave a slack loop in the GPS lead so the cover lifts off. With the cover off you reach the battery, its XT30 and the FC's USB-C socket (left side, beside the GPS lead): plug the laptop in there for setup, parameters and logs.



FOAM TAPE

FUSELAGE ASSEMBLY

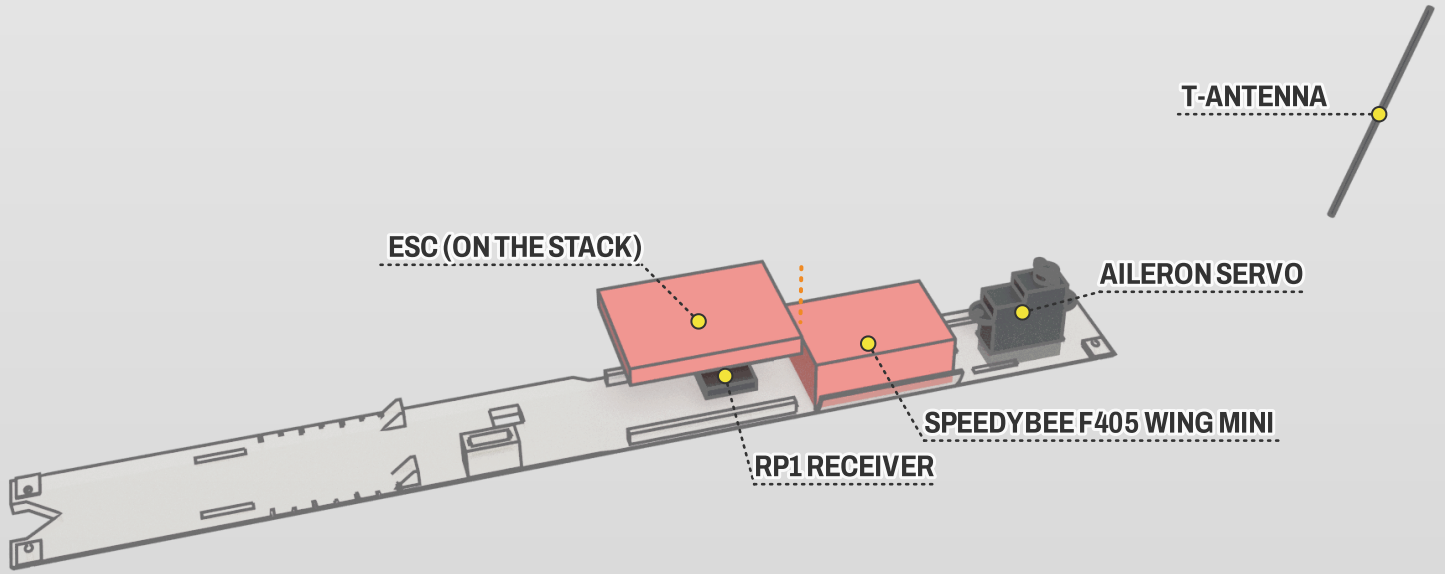
Lower the pack through the hatch against its rear stop (two ribs on the floor), then slide it forward under the tube until its rear edge is at the tick the balance table gives (ticks every 5 mm on the floor edges); one strap through the slots. The XT30 pair lies behind the pack under the GPS and unplugs through the hatch. To take the pack out, slide it back to the stop first.



VELCRO STRAP

EQUIPMENT

SpeedyBee F405 WING MINI stack on foam tape and a strap on the tray's wing-bay floor, arrow forward, servo BEC jumper at 6 V; the ESC on top of the stack on a foam pad. Receiver ahead of it, its T-antenna as an L at x≈220 (one leg up the left wall, one across the floor). Aileron servo aft of the stack, output at the aft end.

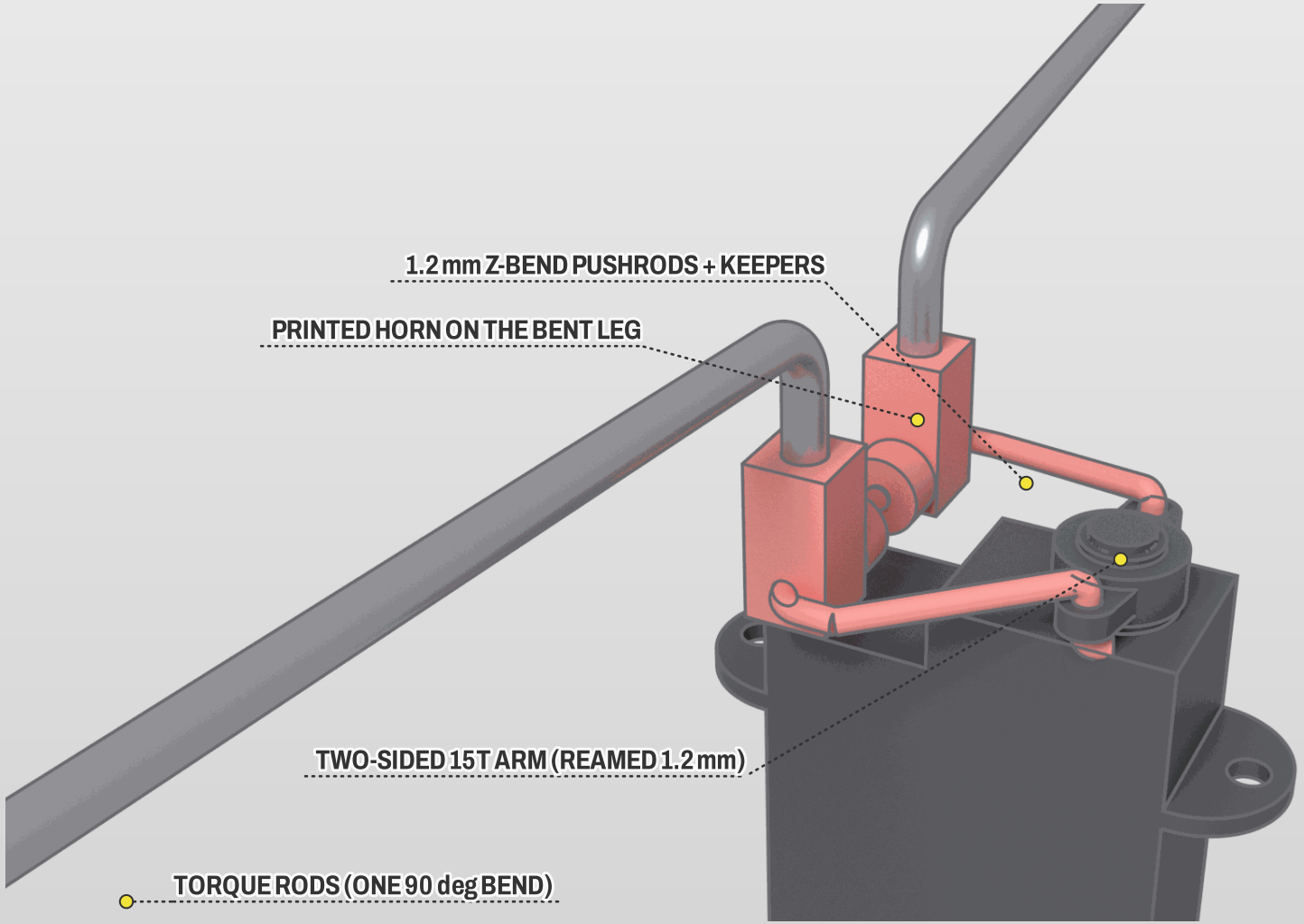


FOAM TAPE

EQUIPMENT

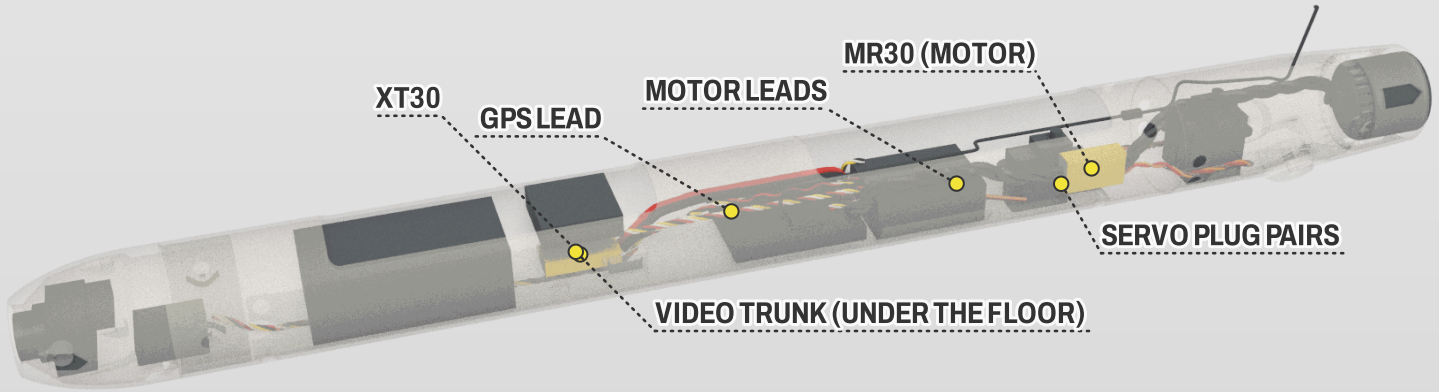
ES9052MD aileron servo standing through the wing-bay floor on foam tape; two-sided 15T arm. Each torque rod has one 90° bend (the horn leg, 6 mm from the centreline); epoxy the printed horn (AHR / AHL) on the leg, 4 mm deep. Drill the arm's 4.3 mm holes 1.1 mm and ream them by hand with a 1.2 mm bit (and both horns' holes): the wire must turn with no play. Each horn connects to the arm with a 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend pushed inboard through the horn's reamed hole, snap-on keeper on the protruding end. Check: with the servo powered, each aileron trailing edge moves <= 0.2 mm (free play lowers aileron flutter to about 33 m/s). Set both ailerons 3° trailing edge up at servo neutral. No play at either end.

Z-BENDS



EQUIPMENT

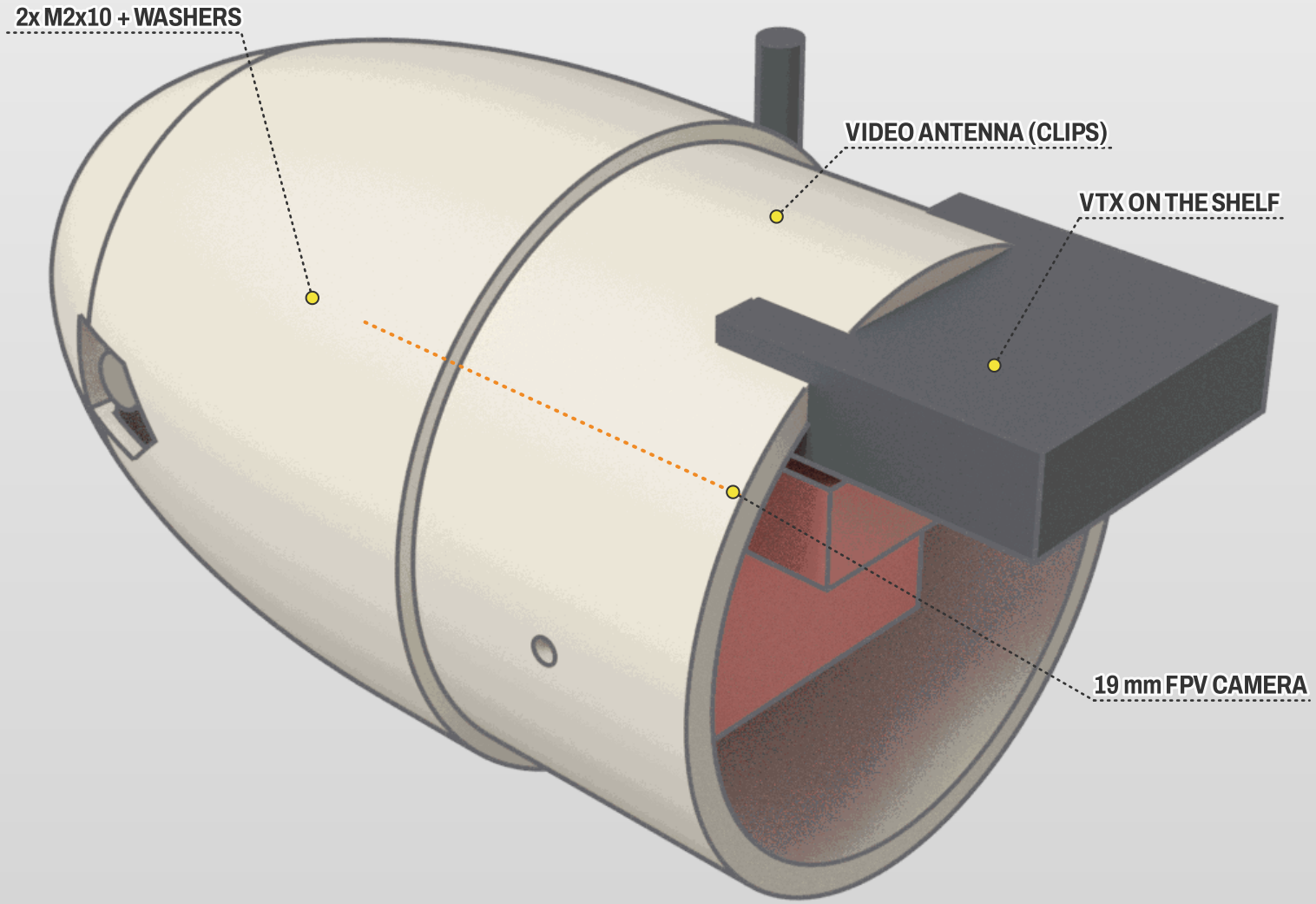
ESC flat on its foam tape at the bottom of the wing bay, receiver on top of it, FC stack 5 mm behind it (servo header row on the right). Battery lead (14 AWG) from the XT30 over the ESC and down the gap to the FC power board's BAT pads; 14 AWG ESC feed from its ESC pads up onto the ESC; ESC signal (M1, GND, TLM) and receiver lead up the gap to the FC; motor phases up the gap, over the FC's left half to the MR30; the ELRS coax under the wing-nut bars to the u.FL joint ahead of the tail joint and up into the fin. Cut lengths and lanes: Electrical Integration manual. Unplug the MR30, the u.FL joint and the servo plugs before pulling the tail module.



HEAT SHRINK

CAMERA NOSE

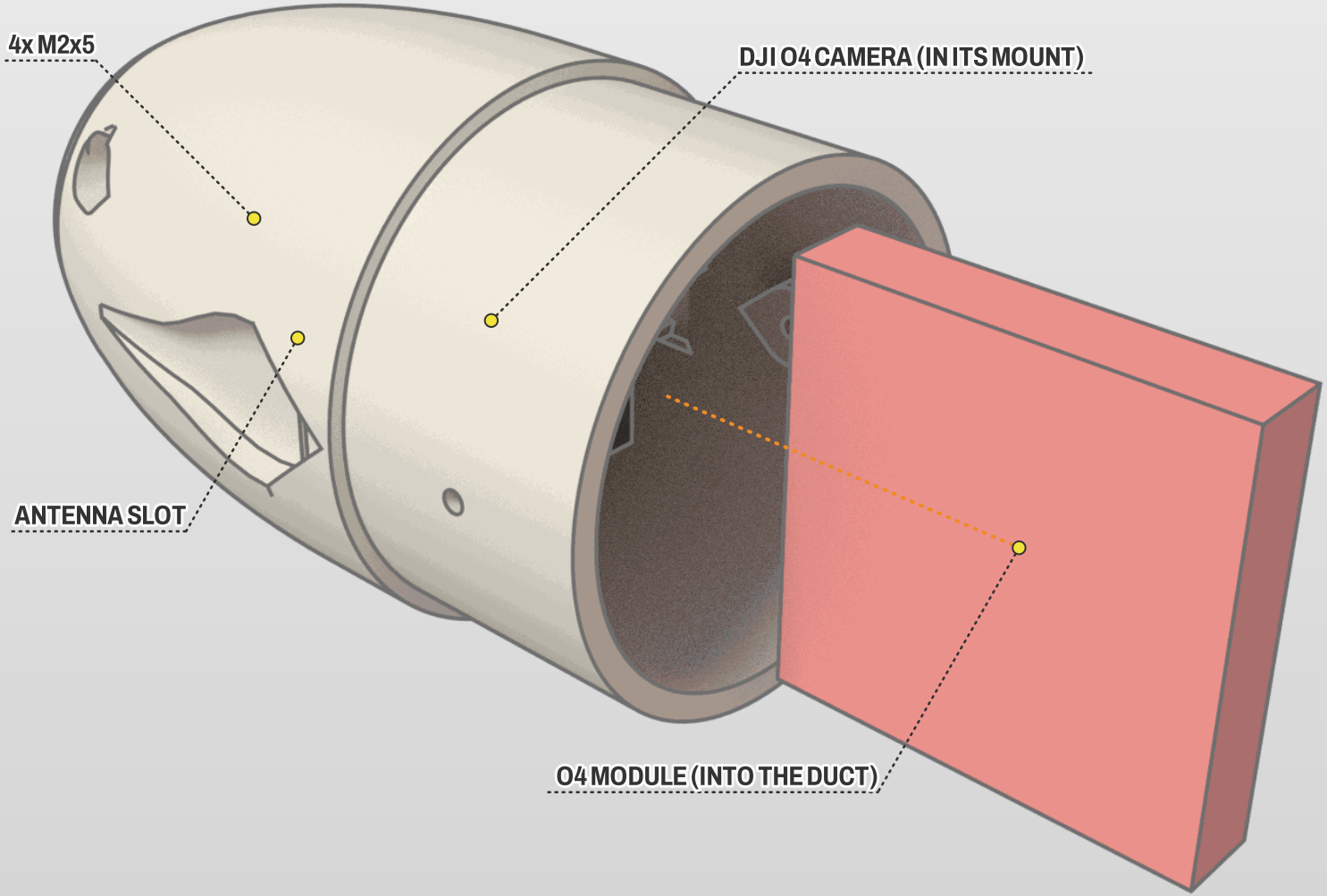
Slide the 19 mm camera in from the open rear until the lens reaches the face; 2 × M2 × 8 pan head + 2 × M2 washer (5 mm OD) through the side seats, lens flush, tilt 0–10° up. Fix the VTX to the printed VTX shelf (foam tape + zip tie through its slots), slide the shelf in from the rear onto its two rails and glue it; video antenna in the two clips on the right wall. Plug the one lead at the nose joint. Dome noses: glue the printed dome ring 3 mm into the front of the tube, then fit the clear dome over its lip with canopy glue (no CA).



FOAM TAPE

CAMERA NOSE

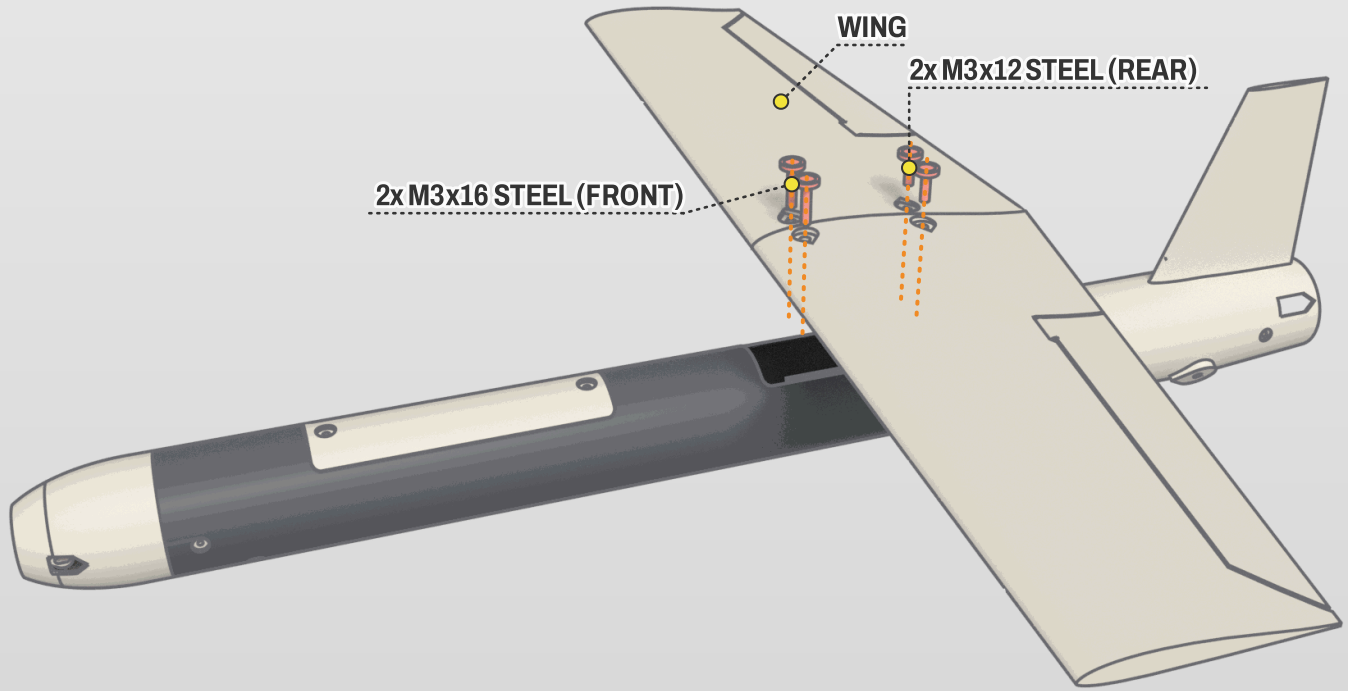
DJI O4 option, in this order: (1) snap the video antenna into its two clips on the left wall, from the rear; (2) slide the camera in its mount in from the back and clamp it with 4 × M2 × 5 through the side slots (at most 3.1 mm into the mount), lens flush; (3) plug the coax and the 3-in-1 lead into the module and slide the module edge-first into the duct insert (DT) until it stops, foam strip in the top groove; (4) slide the insert in through the spigot until its two notches stop on the joint-screw bosses. The tray holds it once the nose is on the tube. Cooling air comes in through the belly NACA and rises over the insert's sill before it reaches the module: never tape the module's faces.



M2 × 5 SCREWS

FINAL ASSEMBLY

Glue the printed window filler (WF) under the wing root (epoxy), centred, with the wing on the tube so it lines up with the window. Push an M3 nut up into each of the bonded nut bars' pockets (two per bar). Set the wing into the tube's rectangular window (the filler plugs in) and bolt it with the steel countersunk screws in steel cup washers (flush in the recesses), two at each station, one in each wing half (each half is clamped on its own, off the glued seam): M3 × 16 at the front, M3 × 12 at the rear; they pull the nut bars up against the tube from inside. Snug them with a small driver, do not overtighten (the nut bars are printed).

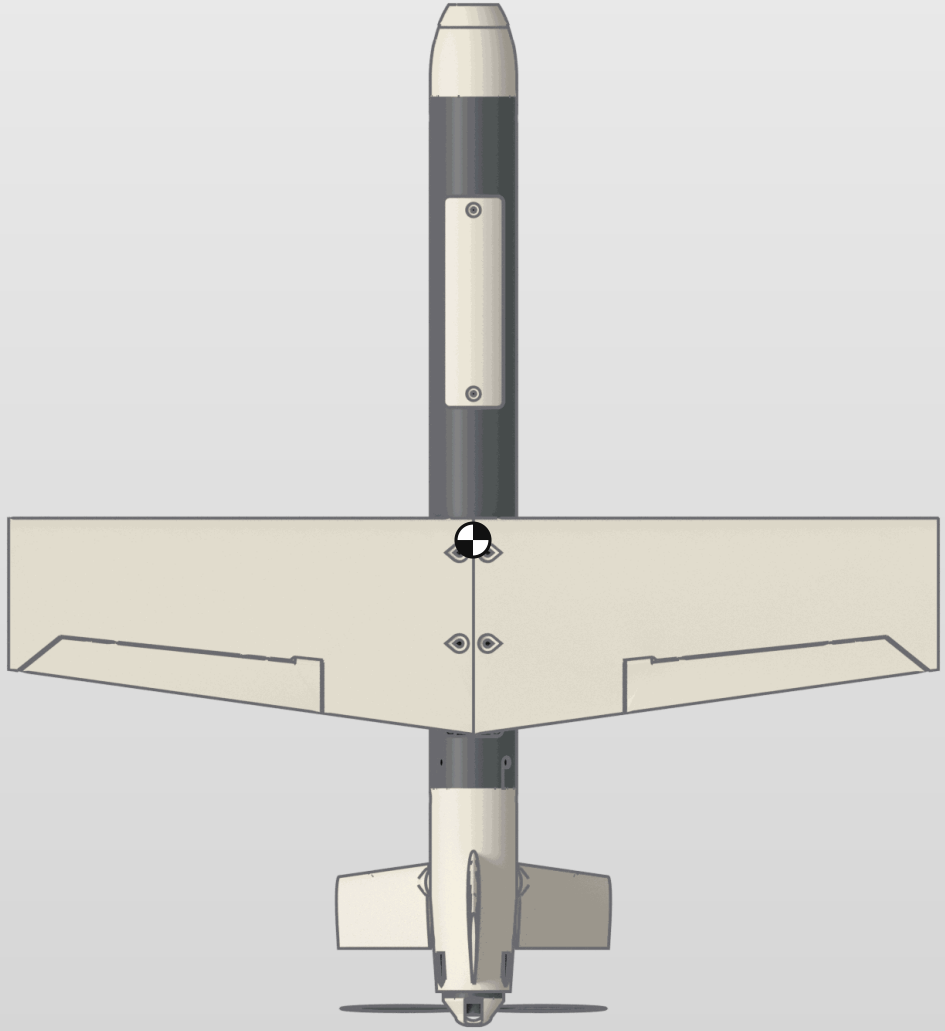


STEEL SCREWS

FINISHING BUILD

Before the first flight: every control surface moves the right way (pitch up: both panel trailing edges up; right roll: right aileron up), throws ±10° on the tail panels, about ±10° (±3.5 mm at the trailing edge) on the ailerons, 30 % expo; the prop turns the right way and pushes aft; failsafe set to motor off.

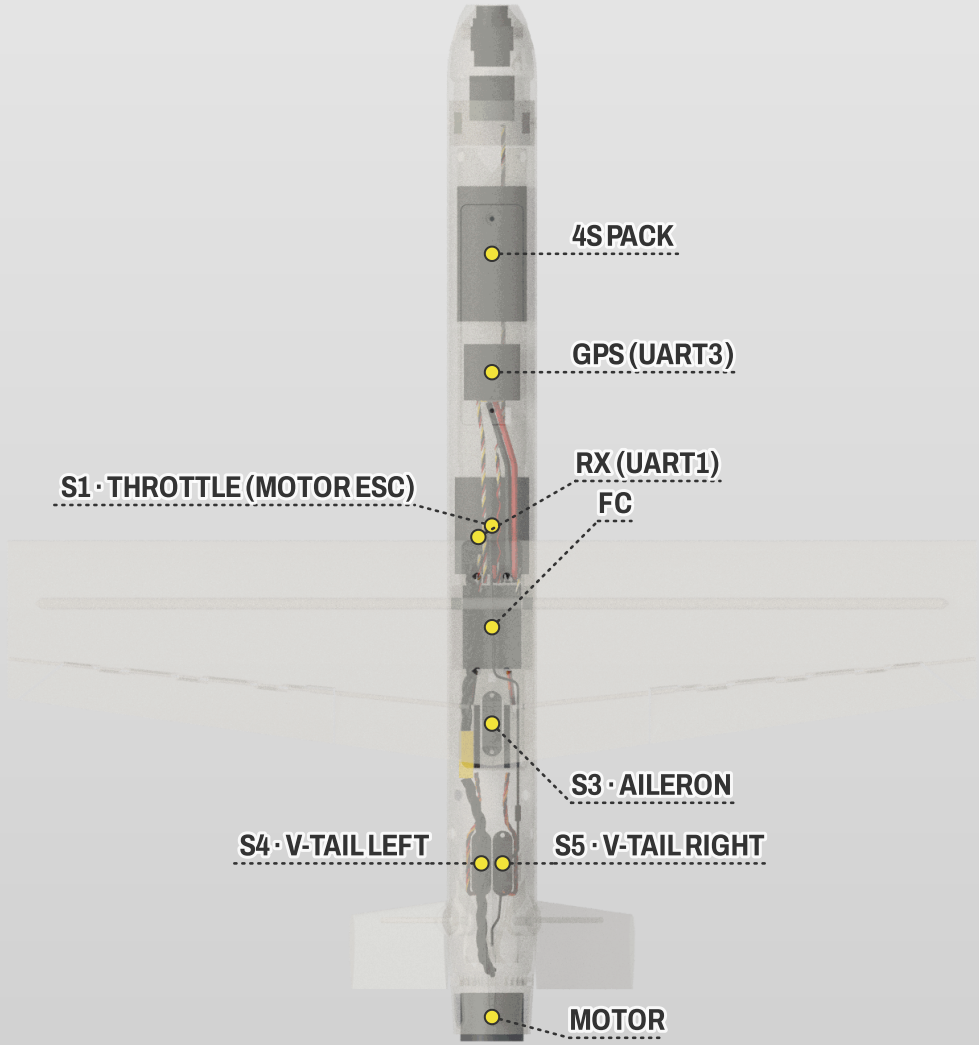
Balance: CG 10 mm behind the wing leading edge at the root (mark it on the wing underside). Slide the battery to trim; for the first flight set the CG 2 mm forward. If a heavy camera needs it, use the nose ballast cup.



WIRING DIAGRAM

The diagram shows the flight-controller outputs and where each device sits. Signal, power and video leads follow the routes on the wiring page of the assembly guide; the package has this diagram as a separate large page.

PAD	FUNCTION	DEVICE
S1	Throttle (motor ESC)	ESC (DShot300, bidirectional)
S2	Unused	Leave empty (DShot group with S1)
S3	Aileron	Aileron servo (ES9052MD, wing bay)
S4	V-tail left	Left tail-panel servo (ES9052MD, tail module)
S5	V-tail right	Right tail-panel servo (ES9052MD, tail module)



ELECTRICAL INTEGRATION

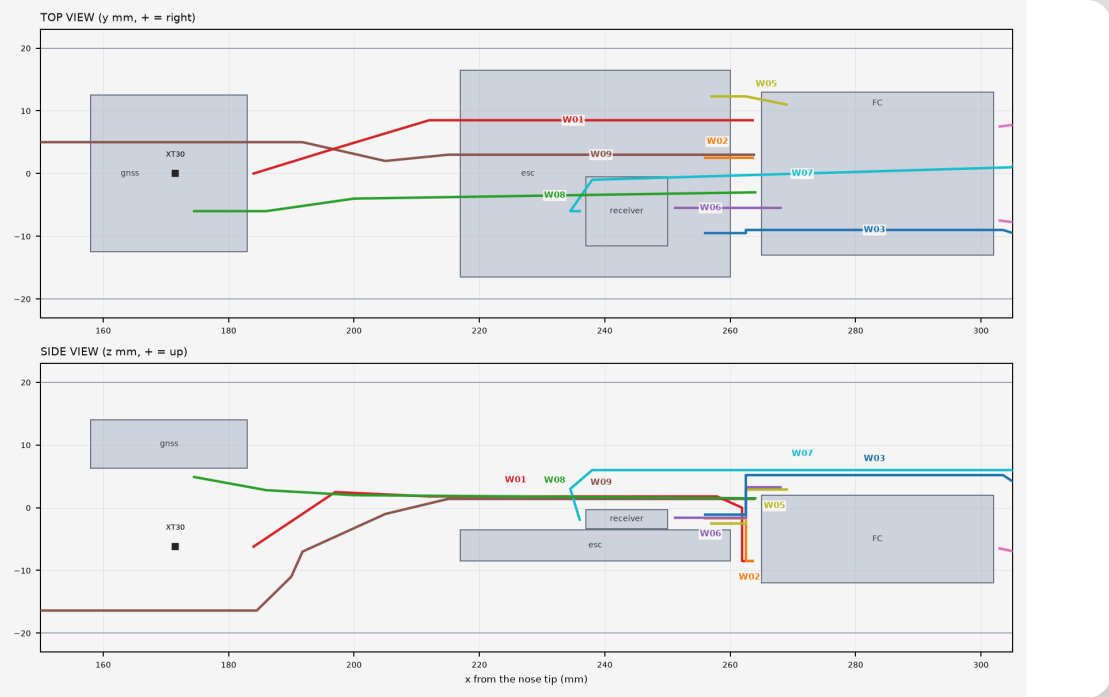
Battery → FC power board (80 A current sensor) → ESC (motor on M1) → MR30 at the tail joint → motor. Main DC path 14 AWG (46 A on a pass, 51 A static), motor phases 16 AWG (38 A RMS). Cable lengths are measured on the 3D harness model; the cut list adds the termination allowances and slack. Full detail, pinouts, settings and tests: the **Electrical Integration manual** (PDF in the package and on the web page).

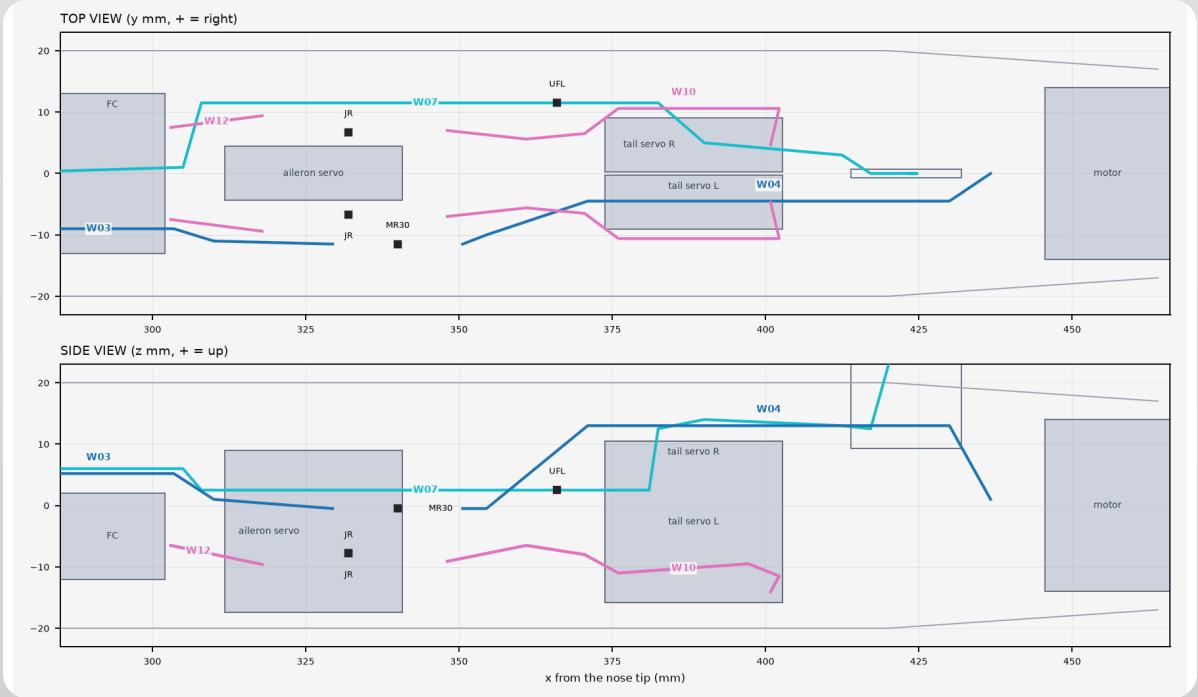
ID	CABLE	FROM	TO	AWG	CUT mm
W01	Battery lead	XT30 female (the pack plugs in), behind the pack under...	FC power board BAT+ / BAT- pads	14	125
W02	ESC feed	FC power board ESC+ / ESC- pads	HDZero Halo 4in1 70A ESC AM32 BAT+ / BAT- pads (XT60...	14	35
W03	Motor phases, ESC side	HDZero Halo 4in1 70A ESC AM32 M1 pads (A, B, C)	MR30 female (ESC side), low on the left ahead of the...	16	110
W04	Motor phases, motor side	MR30 male	Motor phase wires (cut to 20 mm, butt-soldered + heat...	16	140
W05	ESC signal	HDZero Halo 4in1 70A ESC AM32 8-pin SH1.0 lead (only M1,...	FC S1 header (signal, GND); TLM to the R4 pad	28	45
W06	Receiver lead	RadioMaster RP1 V2 pads (5V, GND, TX, RX), on the ESC	FC receiver header (5V, GND, RX1, TX1)	28	45
W07	ELRS antenna coax	RP1 V2 u.FL	Dipole feed at the centre of the fin dipole (z 39.5)	coax	275
W08	GPS lead	TBS M10Q (VCC, GND, TX, RX, SDA, SCL)	FC GPS interface with compass (4V5, GND, RX3, TX3, SDA,...	28	130
W09	Video trunk (N0M nose)	Nose joint plug (camera on the nose; VTX on the nose or,...	FC pads: Vv, G, C1 (camera in), VTX (video out)	28	275
W10	Tail servo leads (L, R)	Emax ES9052MD in F3 (its stock lead)	JR plug pair lying flat beside the aileron servo	26	stock
W12	Tail servo extensions (L, R)	JR plug pair (beside the aileron servo)	FC S4 (left) / S5 (right)	26	stock
W11	Aileron servo lead	Emax ES9052MD lead (stock), aileron servo in the wing bay	FC S3 header	26	stock

ELECTRICAL: POWER AND ROUTING

CABLE	CIRCUIT	CURRENT A (PASS / STATIC)	WIRE	DROP V	dT K / PASS	CONNECTOR RATING
W01+W02	Battery to ESC (DC)	46 / 51	14 AWG	0.08 (0.6 %)	20	XT30: 15 A cont / 30 A peak (see open item)
W03+W04	Motor phases (per phase, RMS)	38 / 42	16 AWG	0.18 (1.2 %)	34	MR30: 30 A cont / 40 A for 1 min per contact

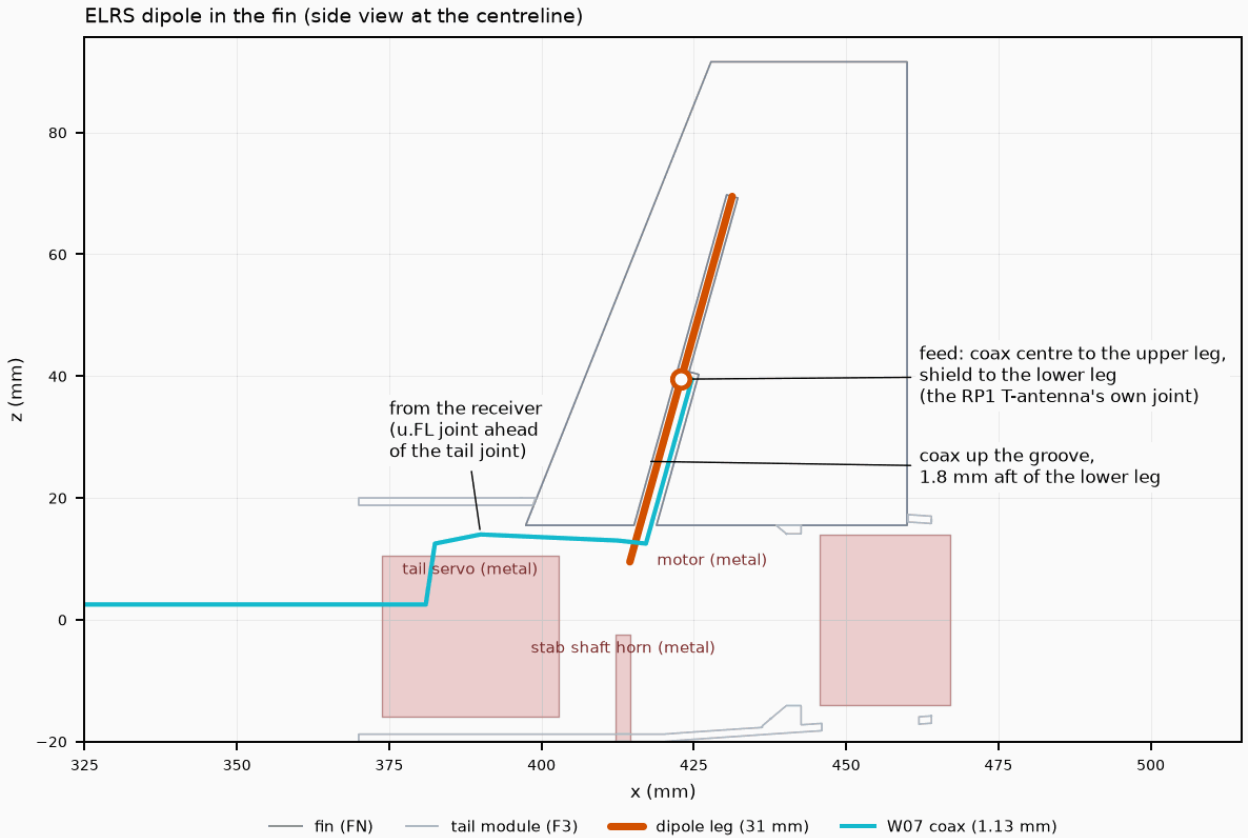
XT30: rated 15 A continuous / 30 A peak. Flown as the test base (70 % throttle cap, no governor) the pack draws about 18 A: fine. Only with the governor fitted (46–51 A) does it need an XT60 lead and a ready-made XT60 pack. Never connect the ESC lead's VBAT pin to the FC's servo header or a 5 V pad. Power-board jumpers: servo BEC 6 V; Vv battery voltage only if the camera and VTX accept 4S.





ELECTRICAL: RECEIVER ANTENNA

The receiver's antenna is a vertical dipole inside the fin, clear of all metal: the RP1 V2 T-antenna's two 31 mm legs lie in a groove on the fin's 30 % chord line with the T's joint (the feed) at the centre. The antenna's own coax runs down a parallel groove beside the lower leg into the tail module, to an in-line u.FL joint just ahead of the tail joint; a 1.13 mm u.FL extension continues to the receiver on the ESC. Unplug the u.FL joint with the MR30 and the servo plugs before pulling the tail module. Lay antenna and coax in the grooves before the fin is glued; three dots of epoxy, tape over the grooves.



ARDUPILOT CONFIGURATION

Starting ArduPlane configuration for SkyG Speed 90B on the SpeedyBee F405 WING MINI (firmware target SpeedyBeeF405WING, ArduPlane 4.5 or later). Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), then work through the bench checks.

Flight controller: on the wing-bay floor, arrow forward; servo BEC jumper set to 6 V for the ES9052MD servos.

Firmware: ArduPlane 4.5 or later · **Board:** SpeedyBee F405 WING MINI (ArduPilot target: SpeedyBeeF405WING)

The setup document and the starting .param file are in the package (ArduPilot folder). Verify every value on the bench before flight.



FLIGHT TEST AND SPEED STAGES

The speed envelope opens in stages. Fly each stage only after the ground checks and the previous stage pass: thrust-stand run, wing load and twist test, control freeplay ≤ 0.2 mm (≤ 0.05 mm above 45 m/s), aileron hinge stiffness, and the tap test. At each run give short aileron and pitch doublets; any buzz or vibration: throttle to zero, gentle pull-up, land and inspect.

Full procedure, limits and log sheet: the test plan in the Engineering folder.

Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	28 %	24 m/s
35 m/s	32 %	28 m/s
40 m/s	36 %	32 m/s
45 m/s	41 %	37 m/s
50 m/s	45 %	41 m/s
55 m/s	50 %	45 m/s
60 m/s	55 %	50 m/s
65 m/s	59 %	53 m/s
70 m/s	64 %	58 m/s
75 m/s	69 %	62 m/s
80 m/s	70 % (needs the governor)	63 m/s
85 m/s	70 % (needs the governor)	63 m/s
90 m/s	70 % (needs the governor)	63 m/s

SKYG

SKYG SPEED 90B

